Sultanate of Oman

Ministry of Agriculture, Fisheries & Water Resources

Consultancy Services for Feasibility Study and Design of Flood Protection Scheme in North and South Al Batinah Governorates



Hydraulic Design of Stormwater Channels

Block No. 4

October 2026

RENARDET S.A. & Partners CONSULTING ENGINEERS L.L.C.

ARTELIA

Document Control

Item	Detail
Project	Consultancy Services for Feasibility Study and Design of Flood Protection Scheme in North and South Al Batinah Governorates
Client	Ministry of Agriculture, Fisheries Wealth and Water Resources, Sultanate of Oman
Consultant	Renardet S.A. & Partners Consulting Engineers L.L.C., in joint venture with Artelia Muscat
Document	Hydraulic Design of Stormwater Channels, Block No. 4
Revision	00
Date	October 2026
Status	Issued for review

	Name	Signature	Date
Prepared by			
Checked by			
Approved by			

Revision	Date	Description
00	October 2026	First issue

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List of Abbreviations

Term	Meaning
ARF	Area reduction factor
CN	SCS curve number
DEM	Digital elevation model
EAV	Elevation-area-volume curve of a reservoir
ECMWF	European Centre for Medium-Range Weather Forecasts
ESA	European Space Agency
FHWA	Federal Highway Administration (United States)
FP	Flood protection (channel or dyke)
FSL	Full supply level
GCA	Global catchment area
HEC-HMS	Hydrologic Engineering Center, Hydrologic Modeling System
HEC-RAS	Hydrologic Engineering Center, River Analysis System
HW	Headwater depth at a culvert inlet
IPCC	Intergovernmental Panel on Climate Change
ISRIC	International Soil Reference and Information Centre
MAFWR	Ministry of Agriculture, Fisheries Wealth and Water Resources
MCM, Mm^3	Million cubic metres
MHHW	Mean higher high water
MHLW	Mean higher low water
MLHW	Mean lower high water
MLLW	Mean lower low water
NDVI	Normalised difference vegetation index
OHDS	Oman Highway Design Standards
OMR	Omani rial
OQ	OQ, the Omani energy company (pipelines)
PMF	Probable maximum flood
PMP	Probable maximum precipitation
Q200	Peak flow of the 200-year event
RUSLE	Revised Universal Soil Loss Equation
SCS	Soil Conservation Service (United States), now NRCS
SDR	Sediment delivery ratio
SM	Stormwater main (channel or dyke)
SS	Stormwater secondary (channel or dyke)
SM2RAIN	Soil-moisture-based rainfall estimate
UTM	Universal Transverse Mercator projection
USBR	United States Bureau of Reclamation
WGS 84	World Geodetic System 1984

Executive Summary

Project context

The Ministry of Agriculture, Fisheries Wealth and Water Resources commissioned Renardet S.A. & Partners Consulting Engineers, in joint venture with Artelia Muscat, to prepare the comprehensive master plan for flood protection dams, river training schemes and stormwater drainage in North and South Al Batinah. Block 04 covers the wadis of Al Hawasinah, Shafan, Sarami and Sakhin between the Hajar foothills and the coast, from Al Khaburah to Saham. The scheme consists of nine flood-protection dams, the flood-protection channels that carry their routed outflows to the sea, and the stormwater channels and dykes that drain the plain between them.

Scope of this report

This report presents the hydraulic design of the 32 stormwater channels of Block 04, 8 main and 24 secondary, and of the 16 stormwater dykes that lead the runoff of the plain to them: the hydrology of the catchments they drain, their alignments and the conditions along them, the design criteria, the section of every channel for the 200-year flow, the outfalls, the culverts at the road crossings, the dyke heights and the cost estimate. The nine dams and the flood-protection channels are presented in the channels feasibility and hydrological report of Block 04.

Key findings

- The 32 stormwater channels total 165.3 km: 68.7 km of main channels and 96.6 km of secondary channels; the 16 stormwater dykes add 13.7 km.
- The channels are sized for the 200-year flow of the plain with the dams in place: the main channels carry 154 to 795 m^3/s with bed widths of 10 to 267 m, the secondary channels 25 to 336 m^3/s with bed widths of 10 to 72 m, at a depth of 2.5 m.
- Eleven of the channels discharge directly to the sea; the others join a flood-protection or a main channel.
- The channels cross the Batinah Highway, the Coastal Road and the planned bypass at 49 points; 637 box-culvert cells of 4 m are proposed, with the full channel depth as clear height. The 16 dykes stand 1.6 to 3.6 m high on the 200-year flood depth of the plain.
- The estimate of the stormwater works is 168.2 million OMR: 47.8 for the construction of the channels, 103.7 for the land of their corridors, 1.8 for the dykes and 14.9 for the culverts, at the unit rates and land rates of the master plan.

1. INTRODUCTION

The study area covers the governorates of North and South Al Batinah in northern Oman. The full project title is “Consultancy Services for Study and Prepare Comprehensive Master Plan for Flood Protection Dams, River Training Schemes and Updating of Flood Maps and Stormwater Drainage System in North and South Al Batinah Governorates, Sultanate of Oman”. The master plan combines dams, dykes and stormwater channels to protect the urbanised areas, the industrial zones, the infrastructure and the planned developments of the plain from flood damage.

Block 04 lies between Al Khaburah and Saham. Its wadis rise in the Hajar foothills and run 15 to 25 km across the coastal plain before reaching the Gulf of Oman. Nine flood-protection dams sit at the foot of the hills and the flood-protection channels take their routed outflows to the sea. Between those channels the plain drains through its own wadis and depressions; the stormwater main channels collect the runoff of the larger intermediate catchments and the stormwater secondary channels drain the local catchments between them. This report covers the 32 stormwater channels and their dykes.

All channels are designed with the upstream dams in place, since the dams attenuate the peak inflows, and the design target is the 200-year flood. The runoff of the plain is routed through the channel network to the sea; the dyke heights follow from the two-dimensional flood simulation. The design process follows the stages of Table 1.

Table 1 – Channel design process – Block 04

Stage	Title	Key components and methods
1	Project input	Design rainfall, topography, land use, dam characteristics (height, reservoir, spillway)
2	Catchment analysis	HEC-HMS subbasin delineation, area reduction factor, SCS curve number, time of concentration, reservoir and reach routing
3	Rainfall-runoff simulation (HEC-HMS)	200-year peak flow and the other return periods at every inflow point of the channels, with the dams in place
4	Channel alignment	HEC-RAS 2D results, wadi geometry, site visits, client and authority input
5	Hydraulic channel design (SewerGEMS)	Manning equation, depth, slope, velocity, freeboard, culvert sizing, constraints (OQ pipelines, bridges and culverts)
6	Performance assessment (HEC-RAS)	Dyke heights, culvert and bridge capacity, outfalls, 200-year protection check
7	Outputs	Channel dimensions, dyke heights, crossing capacities, solutions for bottlenecks

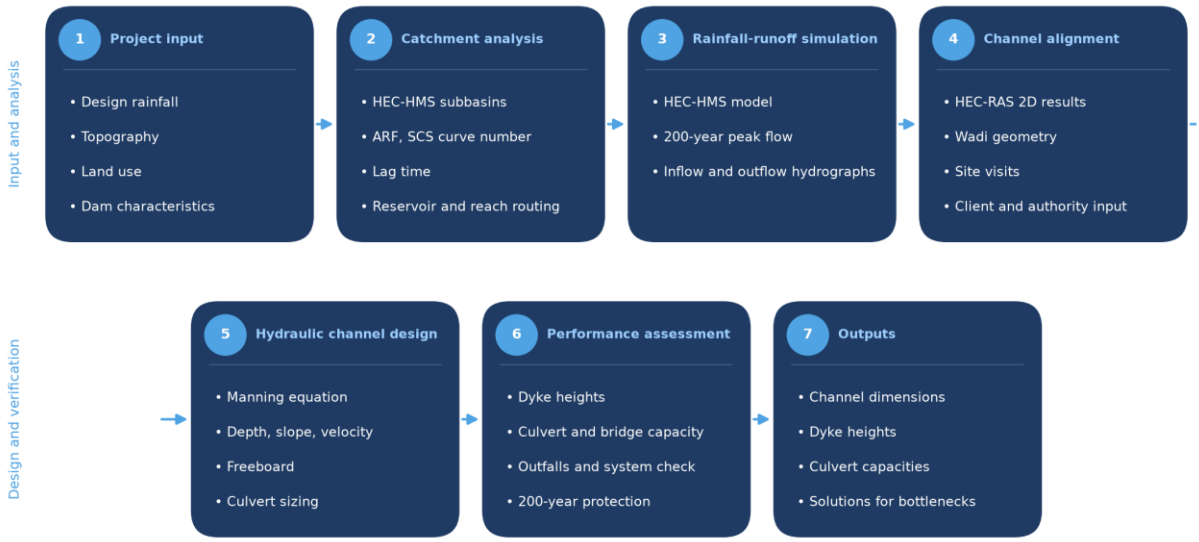


Figure 1 – Channel design process

2. HYDROLOGY

A comprehensive nationwide hydrology analysis was conducted for the Ministry. It covers all the rainfall stations: data cleaning, trend and outlier tests, gap filling, distribution fitting and a regional frequency analysis of the rainfall depth of each region. The 24-hour design precipitation depths and the temporal patterns of that study, for the return periods of 2 to 10 000 years, drive the rainfall-runoff simulation of Block 04.[1]

Table 2 – Design rainfalls

Recurrence interval (years)	Depth (mm)
2	27.6
5	55.3
10	105.5
25	137.5
50	168.1
100	205.4
200	251.0
500	327.2
1000	399.9
2000	488.7
5000	639.1
10000	685.1
PMP	PMP grid, 500 to 1 840 mm

The 24-hour depth is distributed in time with the temporal patterns of the nationwide study (Figure 2), one for the 2-year storm, one for the 5-year, one for the 10- to 1 000-year storms, one for the 2 000- to 10 000-year storms and one for the probable maximum precipitation. All five are centred: the heaviest interval falls at the twelfth hour, and the rarer the event the flatter the pattern, from a 2-year storm that drops a third of its depth in ten minutes to a probable maximum storm spread over the whole day.

[1] MAFWR, nationwide hydrology study: regional frequency analysis of the 24-hour rainfall, design patterns and the PMP grid.

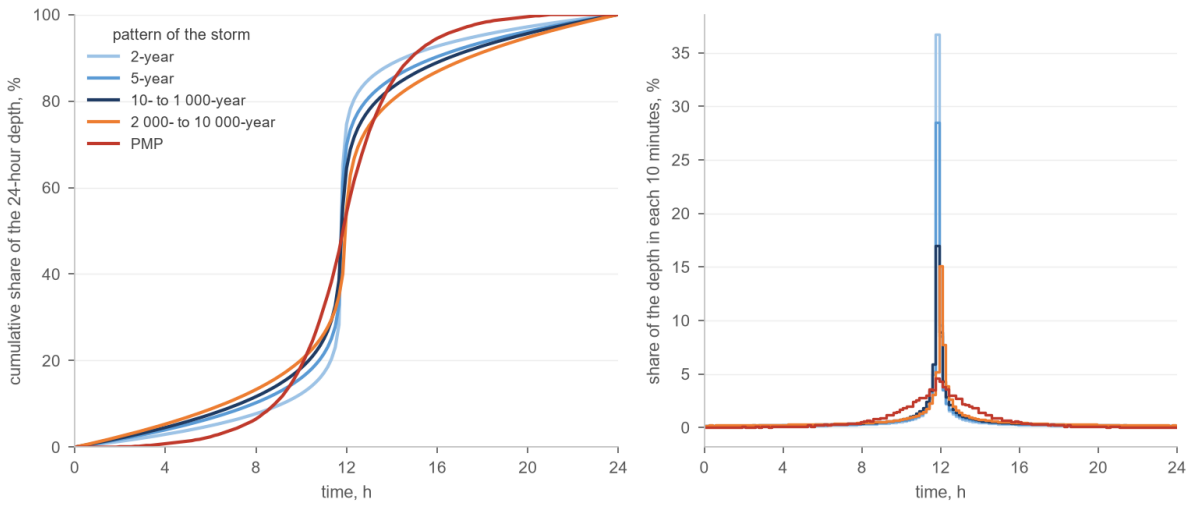


Figure 2 – Temporal distribution of the 24-hour design storm: cumulative share of the depth (left) and the share falling in each 10-minute interval (right), for the five patterns

The probable maximum precipitation is estimated with the storm transposition and moisture maximisation method recommended by the World Meteorological Organisation. The severest historical storms of the region are transposed to the catchment of interest and their precipitable water is raised to its theoretical upper limit on atmospheric thermodynamic principles. The resulting values are the greatest depth of precipitation that is physically possible for a given duration and area, and the basis for the design of critical hydraulic structures. The nationwide PMP grid supplies the 24-hour depth at each catchment; the dam spillways are sized for the probable maximum flood that results from it.

The area reduction factor accounts for the spatial variability of rainfall over a catchment. Point rainfall overstates the average depth over a large catchment, because an intense storm cell seldom covers the whole catchment at once. The factor, a dimensionless value of 1 or less, converts the design point depth to the areal depth. The values of the nationwide study (Table 3) are applied to every catchment through depth-area analyses in HEC-HMS: each element of the model is computed with the storm reduced for its own contributing area, so the peak at a dam site and the peak at a channel junction each carry the right reduction. The reduction matters for the larger catchments of Block 04, where it lowers the computed peaks appreciably.

Table 3 – Area reduction factor

Area (km^2)	ARF
1	1.000
5	0.994
10	0.987
20	0.976
30	0.965
40	0.956
50	0.948
100	0.913
200	0.864
500	0.776
1000	0.694
2000	0.606

The curve number grid combines two global datasets: the ESA WorldCover 2020 land cover at 10 m, which classifies the surface roughness and vegetation, and the ISRIC SoilGrids texture at 250 m, from which the hydrologic soil groups follow. The two are aligned and combined through the standard SCS lookup table into a spatially distributed curve number. The area-weighted curve numbers of the dam catchments lie between 87.7 (Sarami 05 and Sakhin 04) and 90.5 (Hawasinah 11), the response of bare, rocky mountain soils. The initial abstraction and the lag time of each subbasin follow the SCS and Kirpich relations:

$$I_a = 0.2\,S, \quad S = 25.4\left(\frac{1000}{CN} - 10\right) \tag{1}$$

$$t_{lag} = 0.6\,\frac{1}{52}\,L^{0.77}\,S^{-0.385} \tag{2}$$

Ia: initial abstraction, mm

S: potential maximum retention, mm

CN: SCS curve number, –

t_{lag}: lag time, min

L: length of the 10–85 flow path, m

S: slope of the 10–85 flow path, m/m

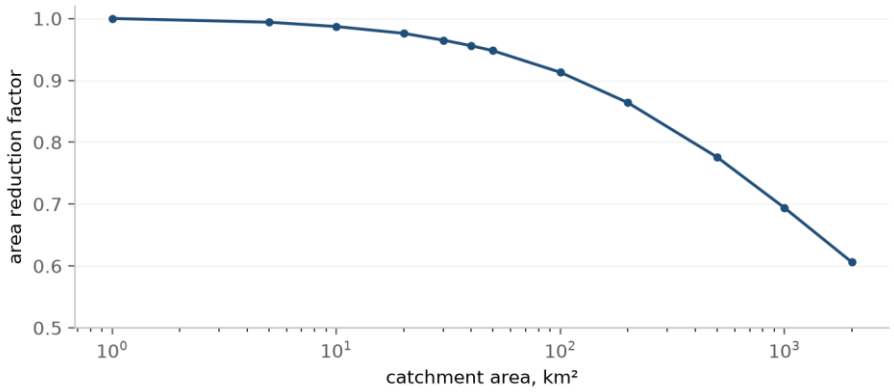


Figure 3 – Area reduction factor against catchment area

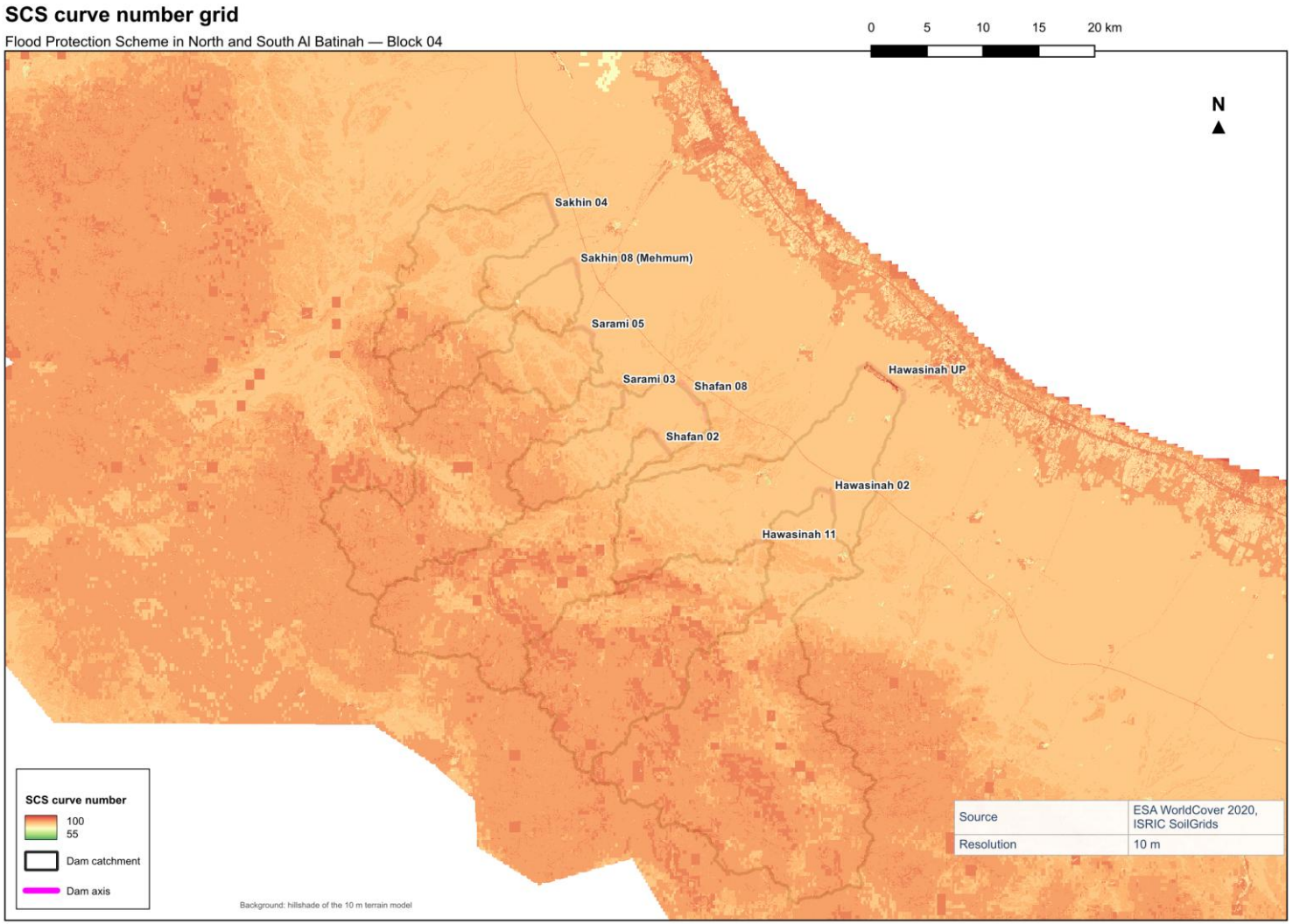


Figure 4 – SCS curve number grid of Block 04

3. CHANNEL ALIGNMENTS

Block 04 covers the four global catchments of Al Hawasinah, Shafan, Sarami and Sakhin (Figure 5), the wadis that reach the coast between Al Khaburah and Saham.

The channel routes were chosen from the two-dimensional flood simulation of the plain, the geometry of the existing wadis, and the site visits made with the client and the local authorities. Where the wadis leave the foothills the flow is confined, and where the plain flattens it spreads and ponds. The channels follow the wadis wherever a defined bed exists and cross the flat ground on the shortest route to the sea.

The stormwater channels come in two classes (Table 4 and Figure 7): the stormwater main channels, which collect the runoff of the larger intermediate catchments of the plain, and the stormwater secondary channels, which drain the local catchments between them. Dykes of the same two classes confine the flow where the ground is too flat for a channel to hold it. The flood-protection channels, drawn in grey on the maps of this report, carry the routed outflows of the dams and are designed in the channels feasibility and hydrological report.

Table 4 – Stormwater channels and dykes in Block 04

Type	Length (km)
Stormwater main channel	69
Stormwater secondary channel	97
Stormwater main dyke	7
Stormwater secondary dyke	7
Total	**179**

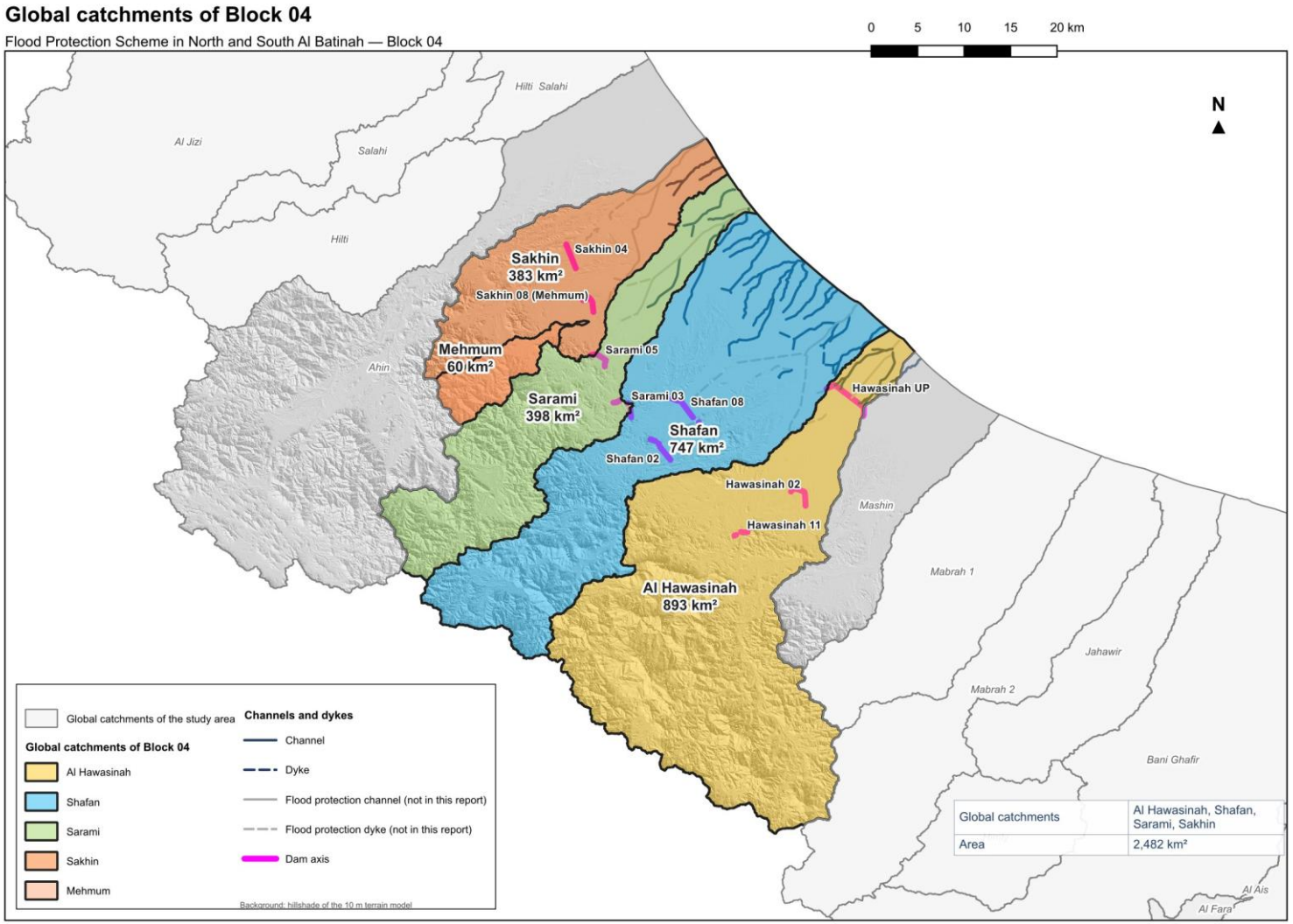


Figure 5 – Global catchments of Block 04: Al Hawasinah, Shafan, Sarami and Sakhin

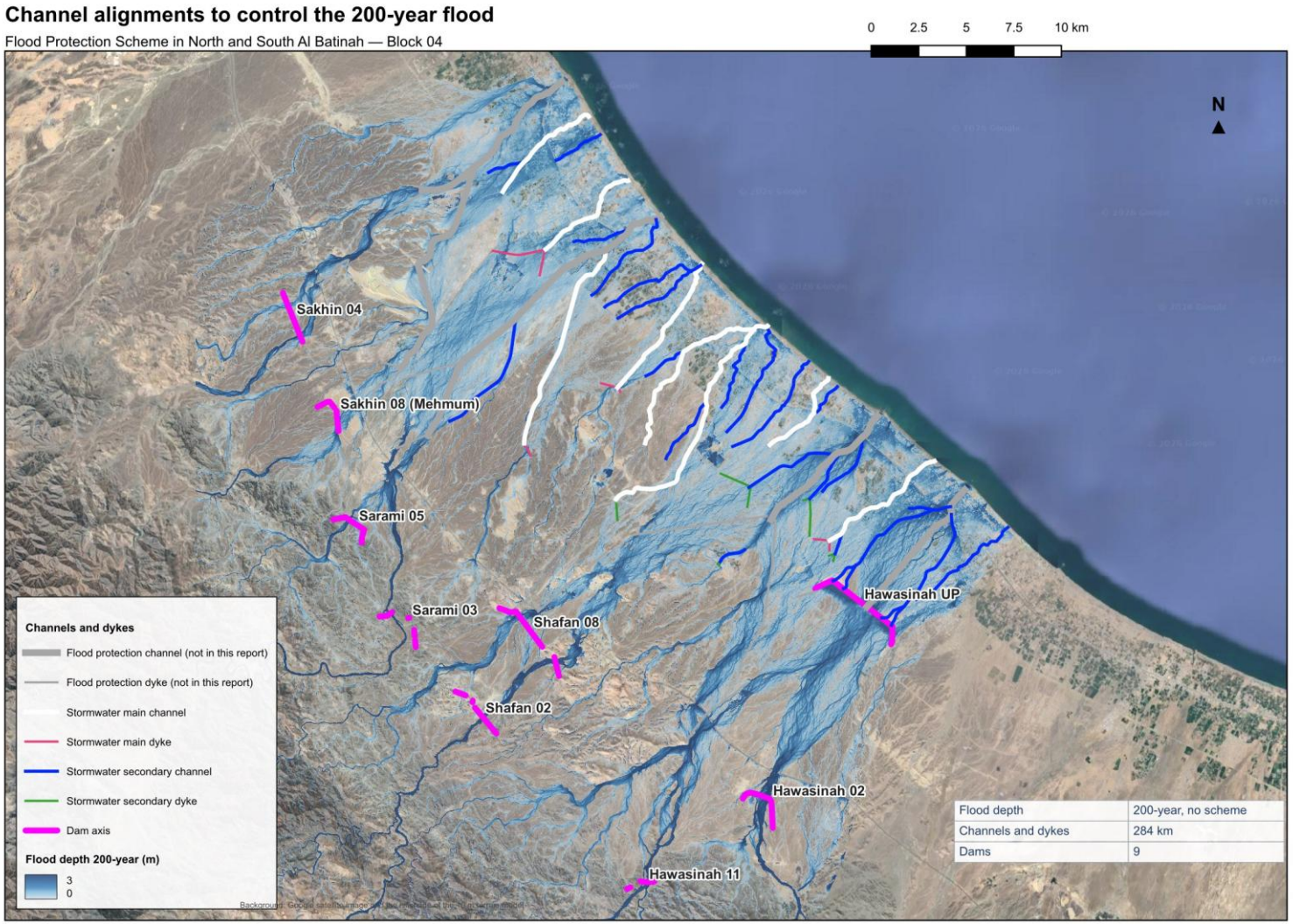


Figure 6 – Channel alignments to control the 200-year flood, on the 200-year flood depth without the scheme

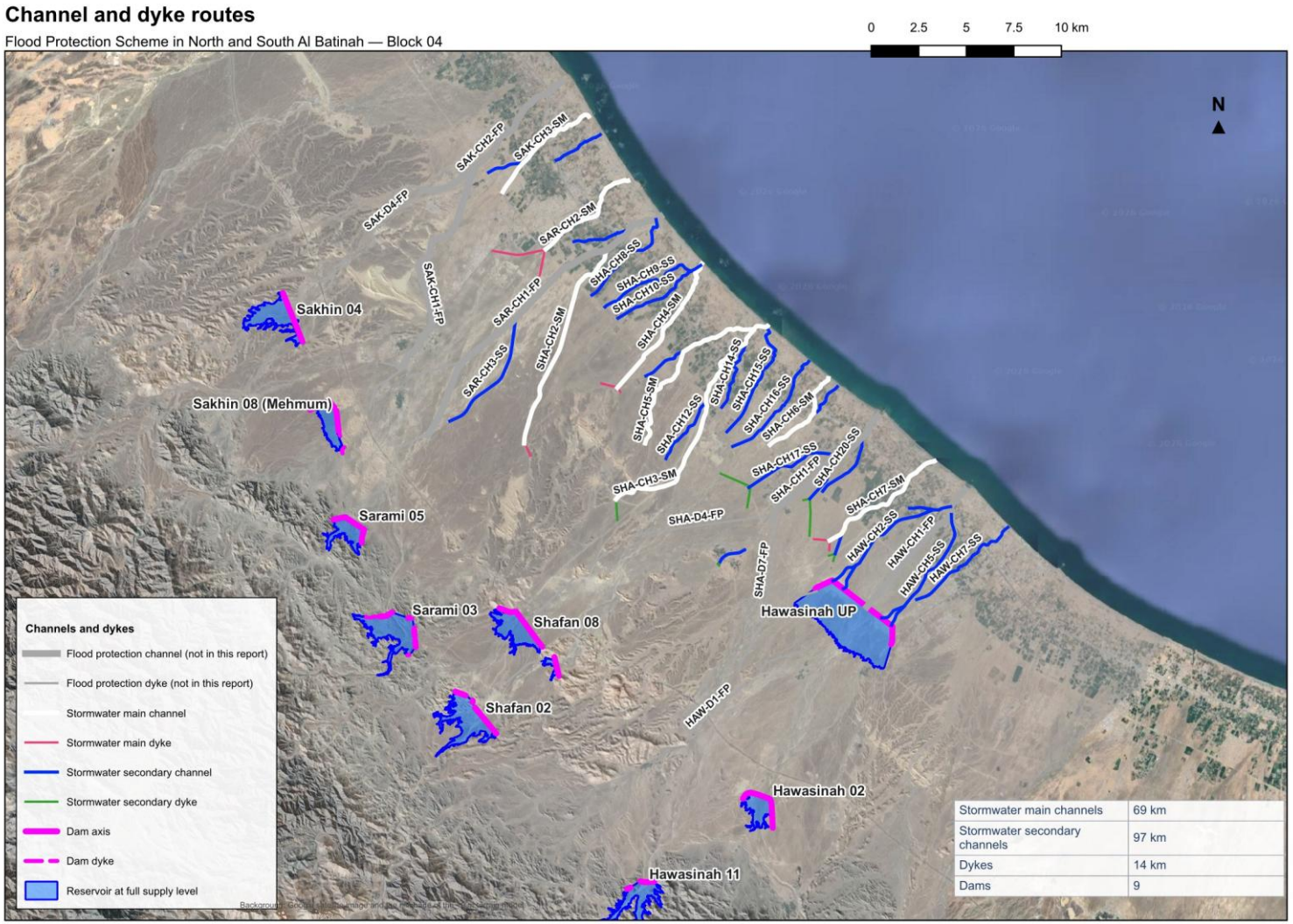


Figure 7 – Stormwater channel and dyke routes by category, with their names; the flood-protection channels in grey

4. CHANNEL CONDITION AND CONFLICTS

Several conditions along the routes govern how the channels can be designed: the existing flow patterns, the roads and the utilities that cross or run along the wadis, and the land use. The flood flow pattern depends strongly on the geometry of the plain; where the wadis leave the hills and spread over the flatter ground, the flow widens and loses its defined bed.

The Batinah Expressway, the Batinah Highway, the Batinah Coastal Road, the planned North Batinah bypass and the OQ pipelines are the main constraints (Figure 8). The existing crossings of the roads fix the channel levels and widths where they are kept, and the shallow pipelines limit the depth of the channels that cross them. Other utility corridors, electricity, water, wastewater and stormwater, also have to be respected and their exact position and depth are required for the detailed design.

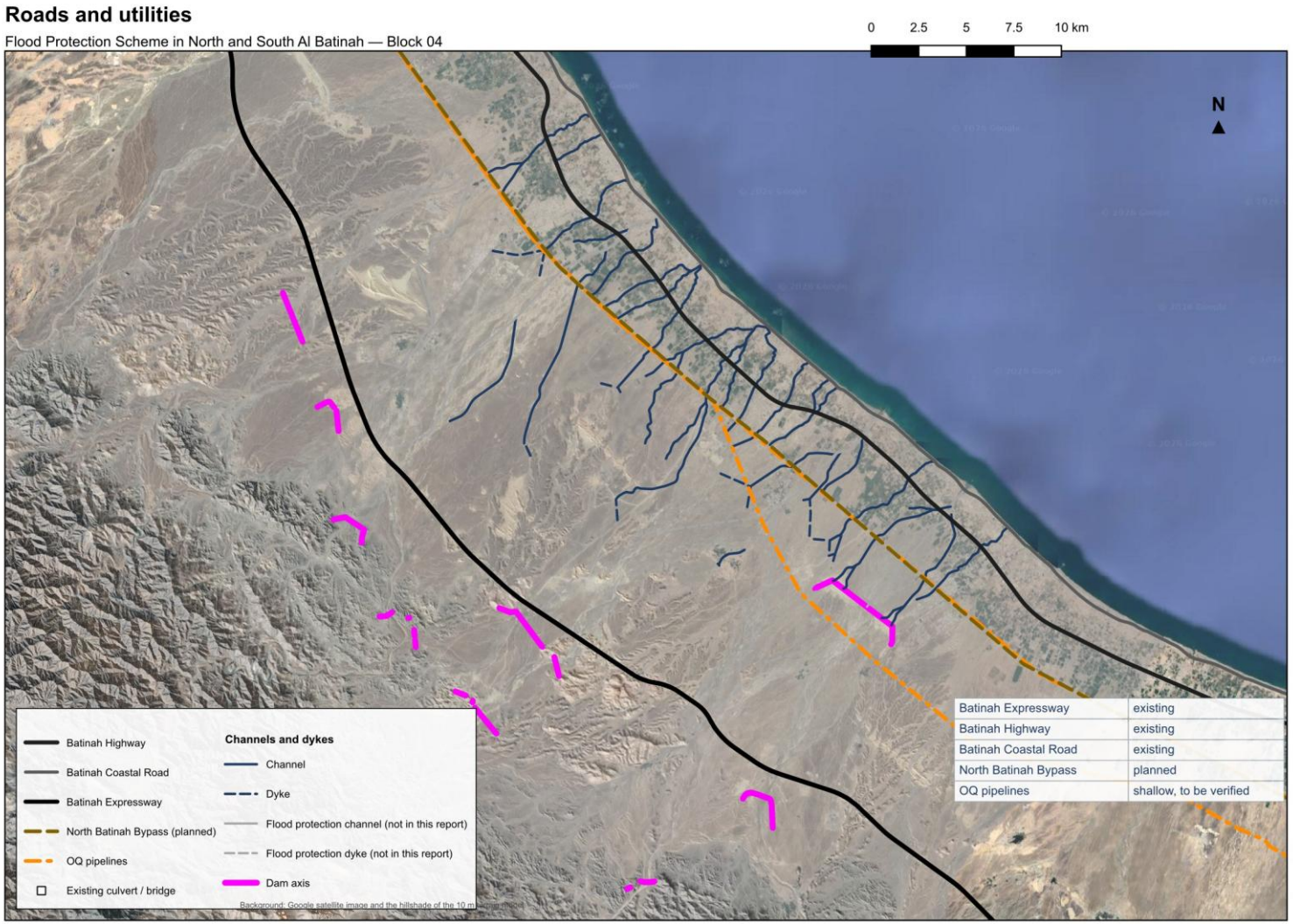


Figure 8 – Roads and utilities

The land use of the plain, managed mainly by the Ministry of Housing and Urban Planning, also shapes the alignments (Figure 9). Houses have been built inside the wadis or have encroached on them and narrowed their beds, and farms extend their boundaries into the wadi. The channels keep to the wadi corridor wherever it is free and deviate around the built plots where it is not.

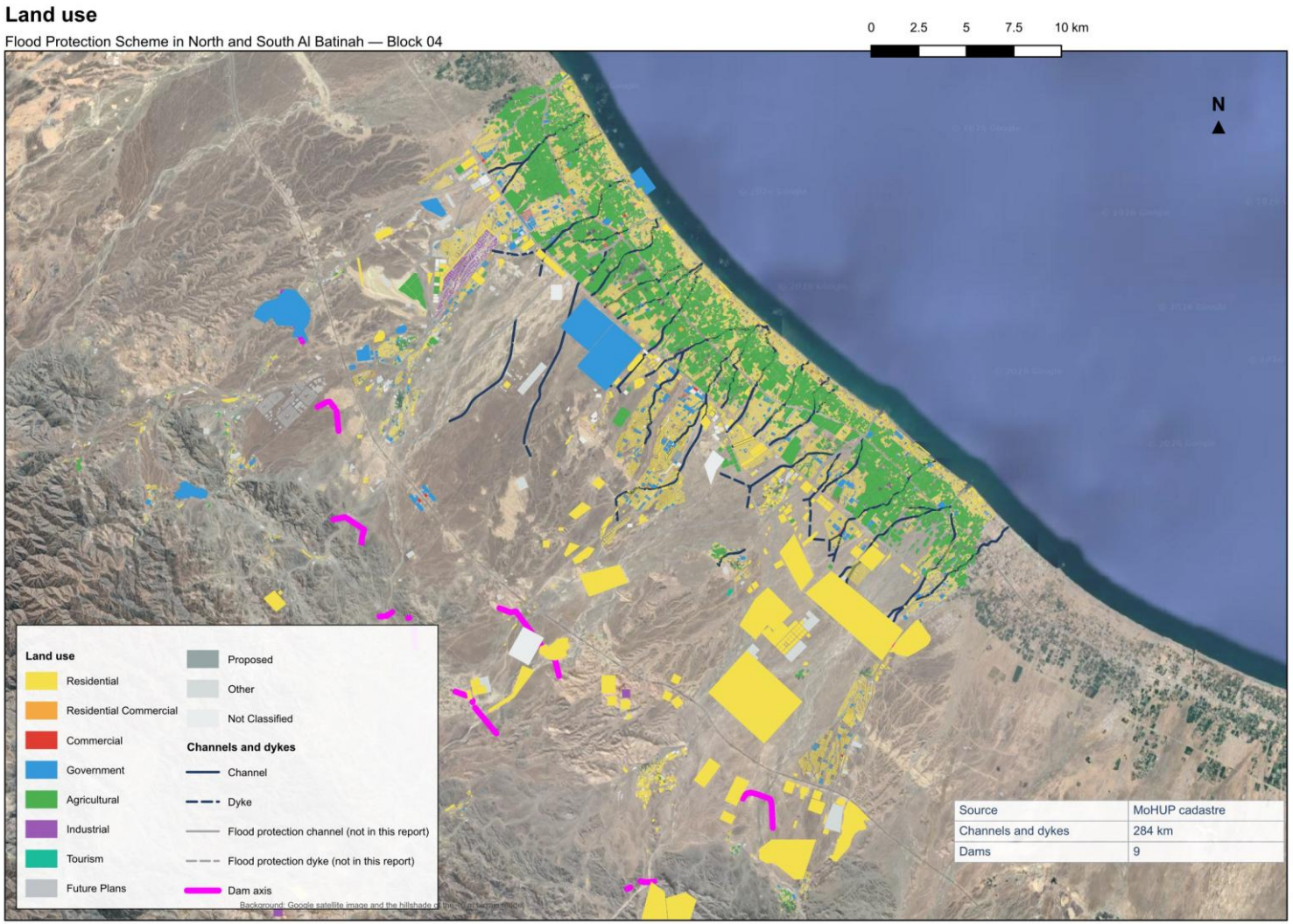


Figure 9 – Land use

5. RAINFALL-RUNOFF SIMULATION

The runoff of the plain is computed with the HEC-HMS model of Block 04 (Figure 10): the catchments of the nine dams with their reservoirs, and the intermediate catchments of the plain between the dams and the sea, each with its SCS curve number and lag time, joined by the reaches of the wadis. The model is run for the design storms of the hydrology study and gives, at every junction, the peak flow and the hydrograph with the dams in place.

Figure 11 and Figure 12 give the 200-year peak flow at the junctions of the model; the flows at the inflow points of the stormwater channels are those of the intermediate catchments, unregulated by any dam, added reach by reach along the channels.

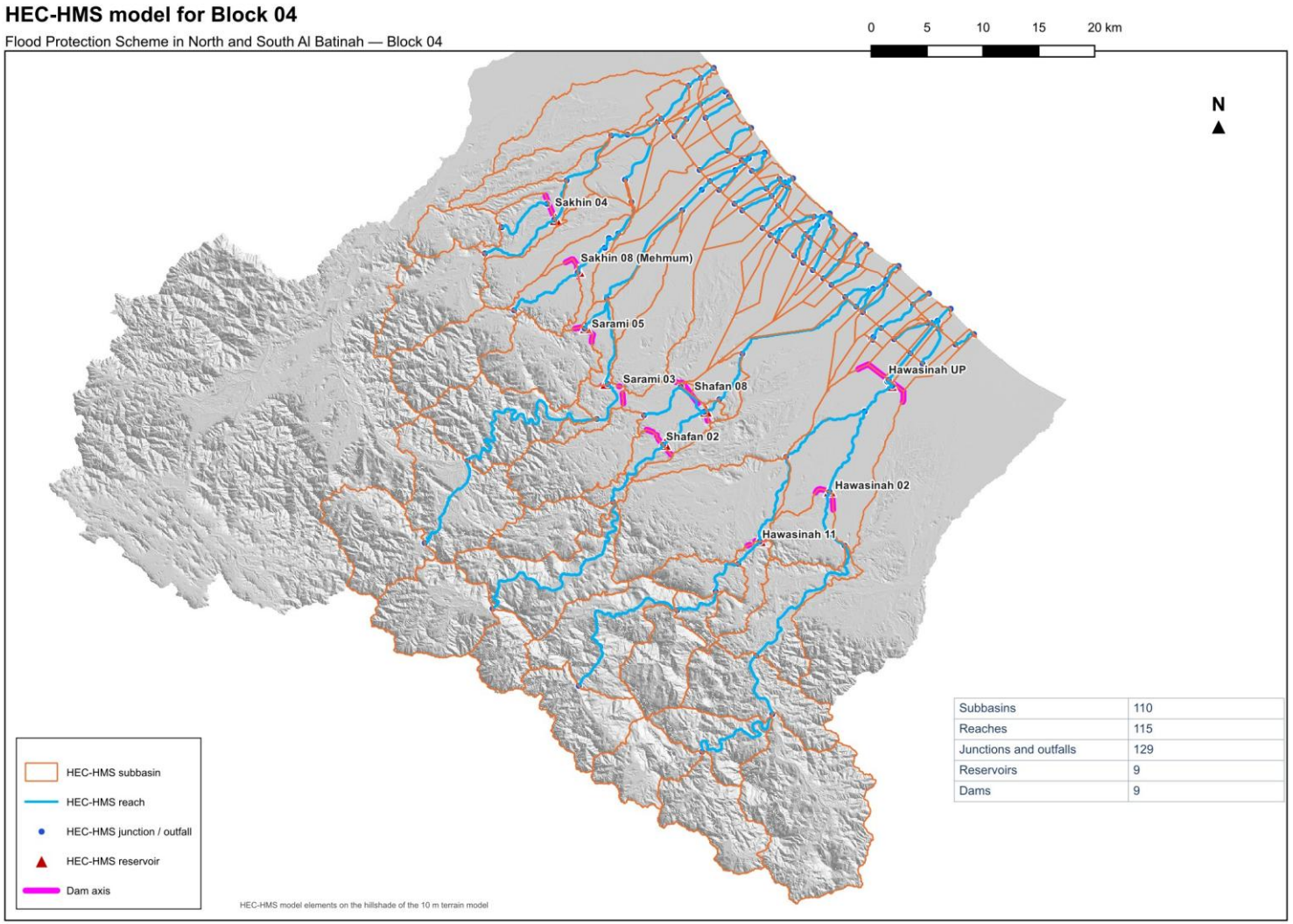


Figure 10 – HEC-HMS model for Block 04

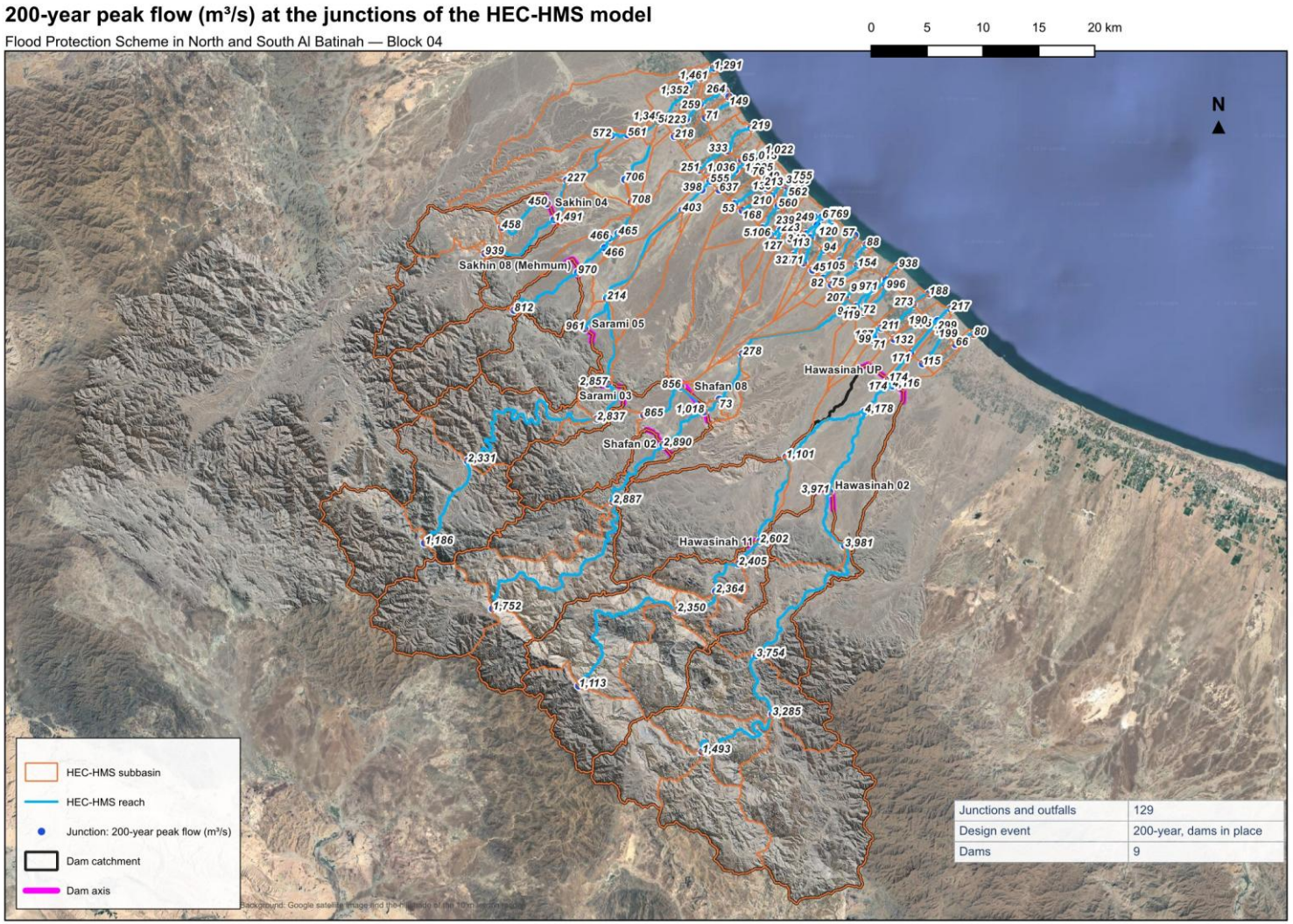


Figure 11 – (a) Q200 (m^3/s) at the junctions of the HEC-HMS model, dams in place

200-year peak flow (m³/s) at the junctions: the dams and the plain

Flood Protection Scheme in North and South Al Batinah — Block 04

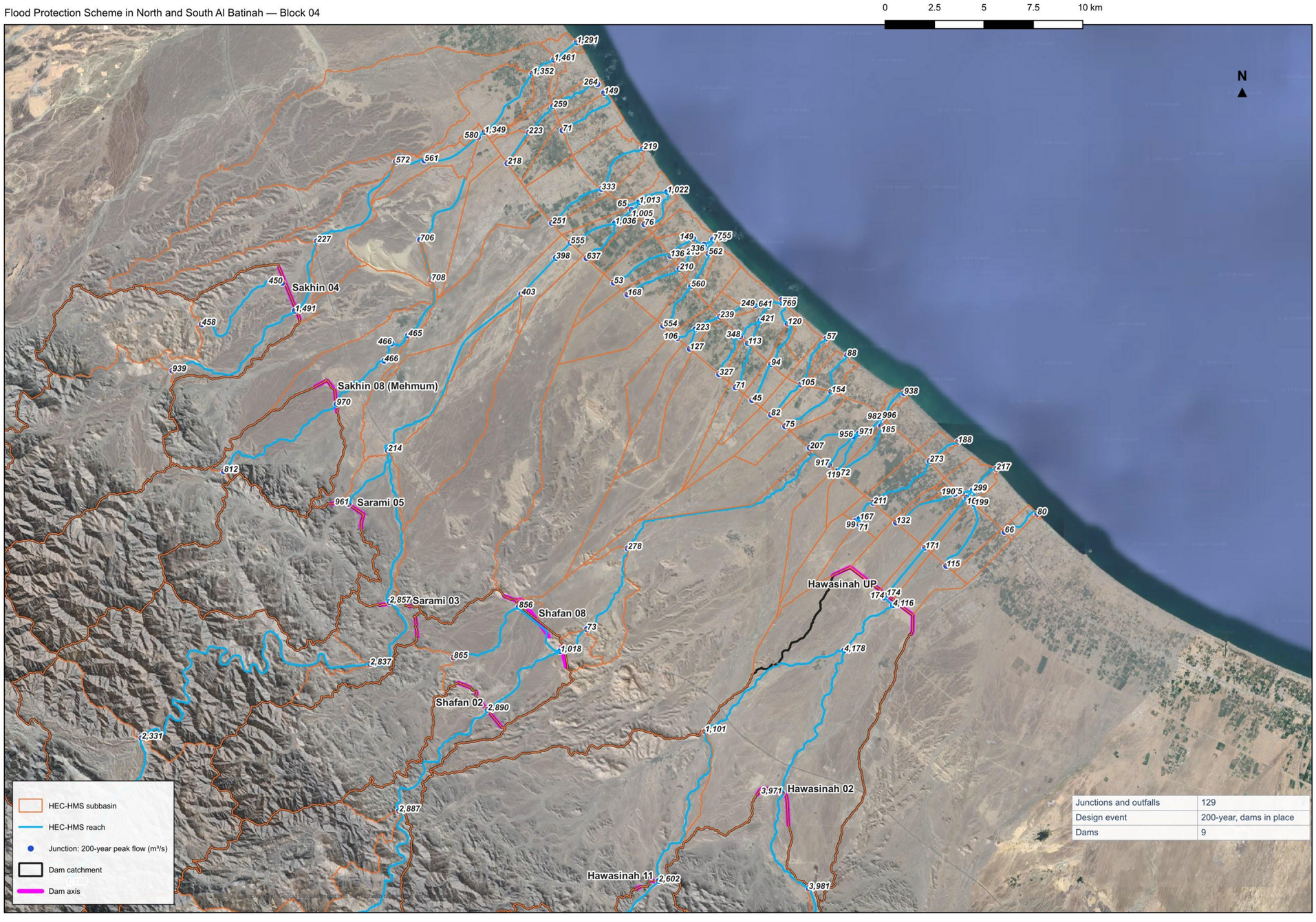


Figure 12 – (b) Q200 (m³/s) at the dams and the junctions downstream of them, dams in place

44 intermediate catchments, 258 km^2 in all, have no dam anywhere in their upstream network: their runoff reaches the channels unregulated (Figure 13). The other 66 catchments, 2,205 km^2, either drain to a dam or receive its routed outflow; their channels are sized on the runoff of the plain added to that outflow.

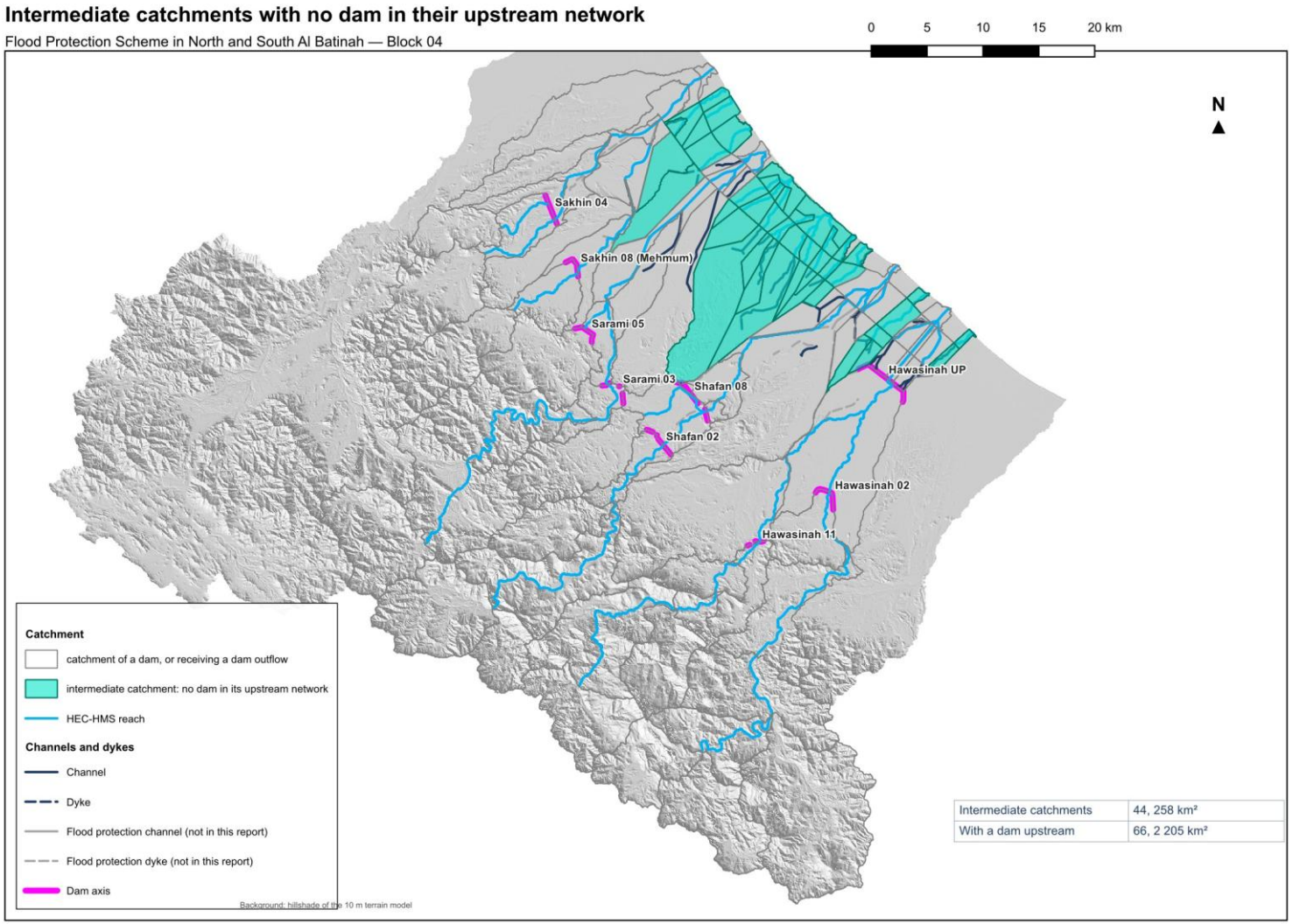


Figure 13 – Intermediate catchments with no dam in their upstream network

6. HYDRAULIC DESIGN CRITERIA

The hydraulic design of the channels follows one set of criteria for the channel geometry, the flow parameters, the slopes, the outfall conditions and the crossings, applied to every reach so that the system is coherent and hydraulically sound. The criteria are summarised in Table 5.

Table 5 – Channel design criteria

Item	Criterion
Channel geometry	Trapezoidal section; depth 2.5 m, or the depth of the existing wadi where it is deeper; side slopes 1:1.5 on both sides; freeboard 0.6 m
Flow formula	Manning equation: $Q = \frac{1}{n} A R^{2/3} S^{1/2}$
Manning's n	Earth channels 0.03; concrete culverts 0.02 (higher than the material value, for sedimentation)
Velocity and slope	Maximum velocity 3 m/s; minimum slope at the flat sea outfalls 0.001 m/m
Outfall conditions	Bank height = max (MHHW, flow depth) + 0.6 m freeboard; the channels terminate at the coastal road crossings
Culvert design	Barrel height = channel height; full channel width covered; slope 0.5 to 5 %; n = 0.02
Cut and fill	Cut preferred, so that the surrounding runoff drains into the channel; fill only where a constraint forces the bed above ground, with embankments on both sides and a return to cut within 500 m
Geometric constraints	OQ pipelines; existing culverts and bridges at the Expressway, the Batinah Highway and the coastal road; outfalls

Table 6 – Mean tidal levels in North and South Al Batinah (m), Oman Maritime Book 2024

Reference port	Mean	MHHW	MLHW	MHLW	MLLW
Al Khaburah	1.80	0.70	0.62	-0.23	-1.00
Saham	1.82	0.68	0.61	-0.25	-1.02
Port of Suhar (Majis)	1.20	1.40	1.30	0.50	-0.30

The channels are trapezoidal in cut and in fill, with side slopes of 1:1.5 on the inner and the outer face. Cut is preferred, because the surrounding surface runoff then drains into the channel; where a constraint forces the bed above the ground, the channel runs in fill between two embankments and returns to cut within 500 m.

The typical sections of the project are shown in Figure 14 for a channel in fill, between two embankments with a 4 m crest, and in Figure 15 for a channel in cut: side slopes of 1:1.5, riprap over geotextile on the inner faces, a toe trench at each bed edge and gravel protection on the outer face of the embankment.

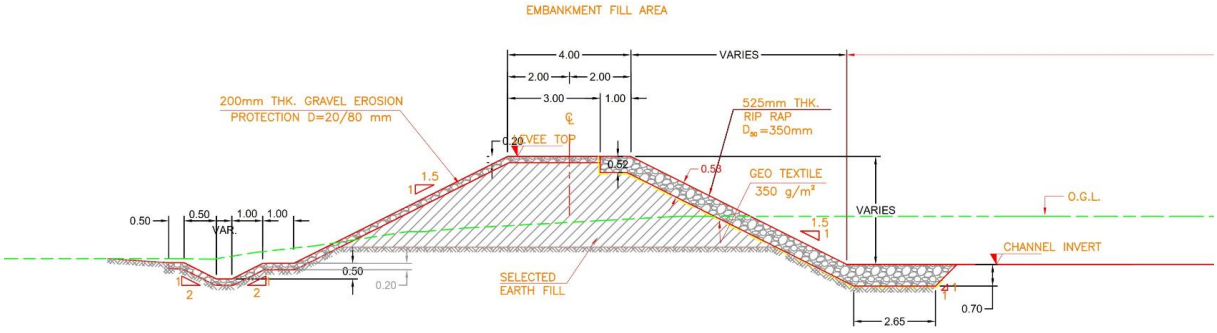


Figure 14 – Typical channel section in fill

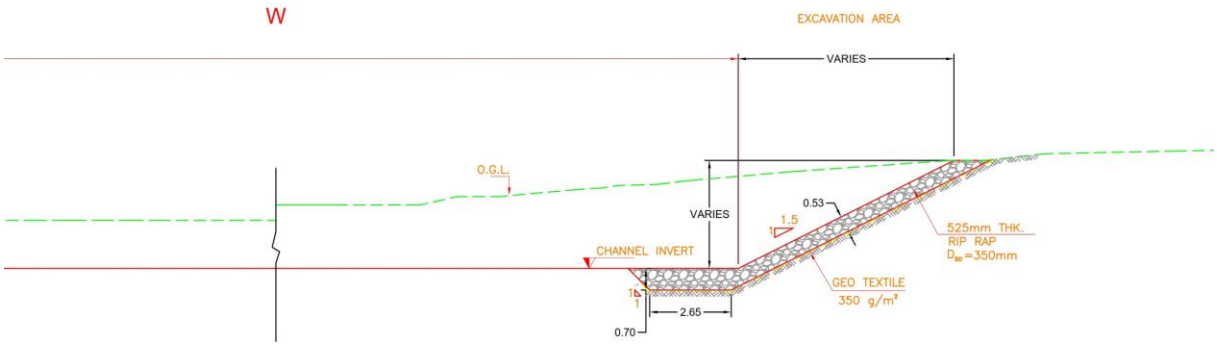


Figure 15 – Typical channel section in cut

Each reach is sized with Manning's equation for uniform flow on the designed bed slope, with the water surface at three quarters of the channel depth:

$$Q = \frac{1}{n} A R^{2/3} S^{1/2} \quad (3)$$

$$A = 0.75\, b\, D, \quad R = \frac{0.75\, b\, D}{b + 1.5\, D} \quad (4)$$

Q: 200-year design flow of the reach, m^3/s

n: Manning's roughness, 0.030, $s/m^{(1}/3)$

A: flow area at three quarters of the depth, m^2

R: hydraulic radius, m

S: bed slope of the reach, m/m

b: bed width, the smallest value in steps of 0.5 m that carries Q, m

D: channel depth, 2.5 m or 3 m, m

At the sea outfalls the bank height follows the tidal level or the flow depth, whichever is higher, plus the freeboard:

$$H_{bank} = max(MHHW, y) + 0.6 \quad (5)$$

H_{bank}: bank height above the channel bed, m

MHHW: mean higher high water of the nearest reference port, m

y: flow depth at the outfall, m

6.1. MAIN ISSUES IN CHANNEL DESIGN

A number of constraints and site-specific challenges affect both the hydraulic performance of the channels and the practicality of their construction. They fall in four groups: road and culvert crossings, energy corridors and utilities, wadi path and land-use conflicts, and sand mining and environmental risks.

Table 7 – Challenges in the design of the channels

Group	Issues
Culverts and road crossings	Existing culvert and bridge sizes insufficient for the channelised flow; drainage crossings that have altered the wadi paths downstream; floodways recommended for the internal roads
Energy corridors and utilities	Conflicts with the OQ pipelines, which are shallow; the exact position and depth of every utility are required for the detailed design
Wadi path and land-use conflicts	Wadi ownership by the Ministry is critical for enforcement; wadis blocked by constructions; depressed areas where the wadis flatten into wide plains and pond around homes and farms; wadis narrowed as farm owners extend their boundaries; the wadi bed to be reserved against developments within the 200-year flood extent
Sand mining and environmental risks	Sand extraction up to 3 m deep destabilises the channel beds; animal carcasses in the ponds after floods are a health hazard
Consequences	With the above constraints not sorted out, the system will not be designed for cyclone-level protection; conveyance is constrained by undersized crossings, narrowed wadis and shallow pipelines; the upstream dams are essential to capture and reduce the flood volumes before they reach the developments

6.2. CHANNEL DEPTH

The design depth is 2.5 m on every reach of these channels (Figure 16); the 3 m depth of the deeper wadis applies to the flood protection channels only. The depth of every reach is in the appendix table.

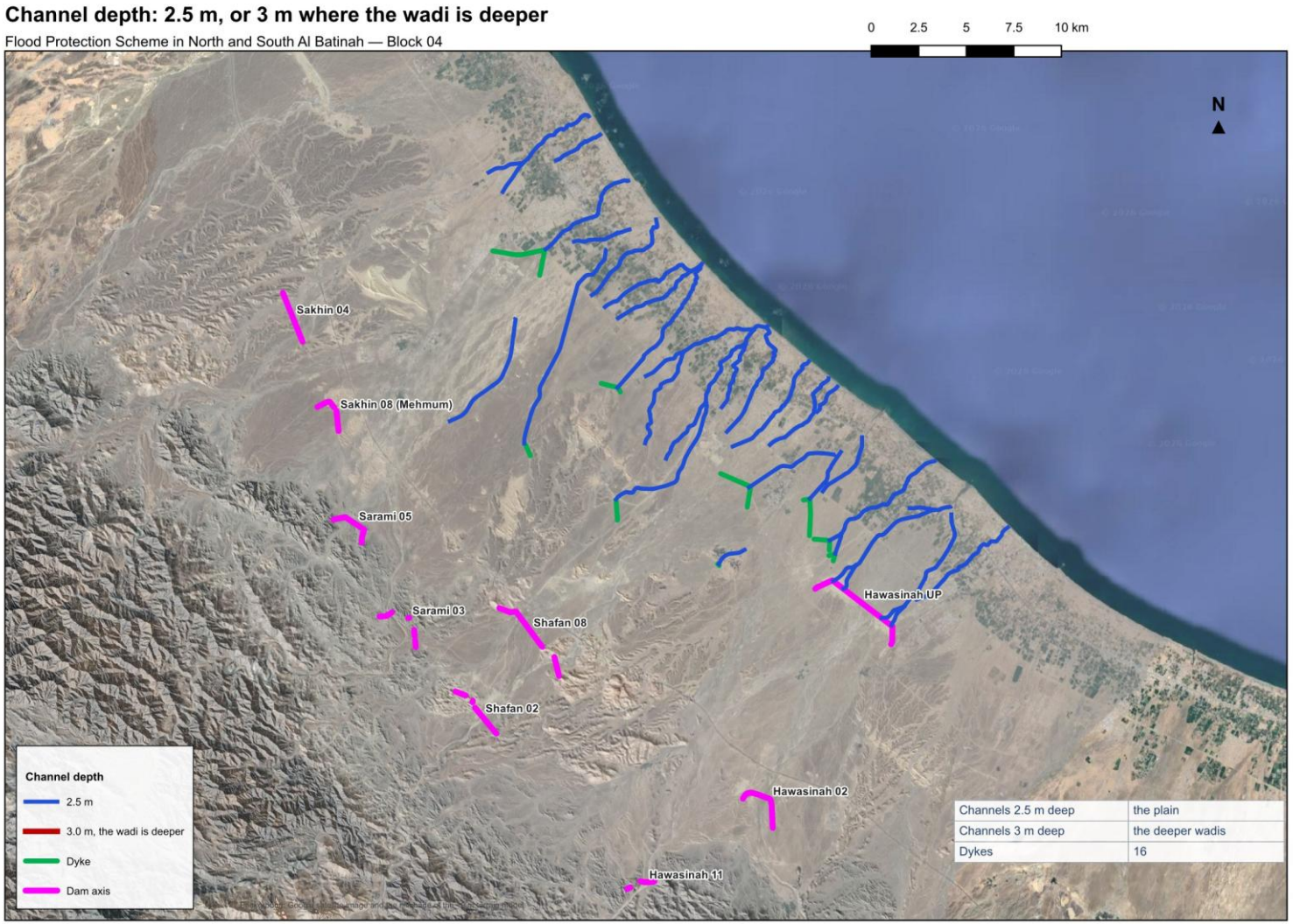


Figure 16 – Channel depth: 2.5 m, or 3 m where the wadi is deeper

6.3. CHANNEL SECTIONS FOR THE 200-YEAR FLOW

Every channel is sized for the 200-year flow it carries with the nine dams in place: the routed outflows of the dams and the runoff of the plain, added reach by reach at the junctions of the hydrological model. Table 8 gives, for each channel, its class and length, the 200-year design flow of its largest reach, the bed width of its first and last reach and the depth of its sections. The sections are sized in the SewerGEMS hydraulic model on the designed bed slope with Manning's n of 0.030, the water at three quarters of the depth and the width in steps of 0.5 m; the width grows from reach to reach as the tributaries join.

Table 8 – Channel sections for the 200-year flow

Channel	Class	Length (km)	Design flow (m^3/s)	Bed width (m), first to last reach	Depth (m)
HAW-CH2-SS	Stormwater secondary	7.7	215	10 to 54	2.5
HAW-CH3-SS	Stormwater secondary	1.4	–	10	2.5
HAW-CH4-SS	Stormwater secondary	1.4	–	10	2.5
HAW-CH5-SS	Stormwater secondary	7.1	199	10 to 44.5	2.5
HAW-CH6-SS	Stormwater secondary	1.3	25	10	2.5
HAW-CH7-SS	Stormwater secondary	6.5	80	12.5 to 29	2.5
SHA-CH2-SM	Stormwater main	11.4	637	83.5 to 107.5	2.5
SHA-CH3-SM	Stormwater main	13.9	795	10 to 267	2.5
SHA-CH4-SM	Stormwater main	8.3	753	94.5 to 253	2.5
SHA-CH5-SM	Stormwater main	9.9	249	20.5 to 50	2.5
SHA-CH6-SM	Stormwater main	5.2	154	14.5 to 46	2.5
SHA-CH7-SM	Stormwater main	7.4	273	15 to 58	2.5
SHA-CH8-SS	Stormwater secondary	5.9	76	13 to 20	2.5
SHA-CH9-SS	Stormwater secondary	5.5	149	10 to 52	2.5
SHA-CH10-SS	Stormwater secondary	5.4	336	25 to 72.5	2.5
SHA-CH11-SS	Stormwater secondary	2.4	106	22	2.5
SHA-CH12-SS	Stormwater secondary	3.8	–	10	2.5
SHA-CH13-SS	Stormwater secondary	1.6	313	49.5	2.5
SHA-CH14-SS	Stormwater secondary	3.9	113	12 to 24	2.5
SHA-CH15-SS	Stormwater secondary	6.8	120	10 to 28.5	2.5
SHA-CH16-SS	Stormwater secondary	6.4	105	14 to 37.5	2.5
SHA-CH17-SS	Stormwater secondary	5.1	207	29 to 42	2.5
SHA-CH18-SS	Stormwater secondary	3.0	119	37	2.5
SHA-CH19-SS	Stormwater secondary	1.8	–	10	2.5
SHA-CH20-SS	Stormwater secondary	3.8	185	12.5 to 40.5	2.5
SHA-CH21-SS	Stormwater secondary	1.1	71	11.5	2.5
SAR-CH2-SM	Stormwater main	6.1	333	49 to 94	2.5
SAR-CH3-SS	Stormwater secondary	6.9	336	41 to 49	2.5
SAR-CH4-SS	Stormwater secondary	3.1	65	15	2.5
SAK-CH3-SM	Stormwater main	6.5	264	45 to 90.5	2.5
SAK-CH4-SS	Stormwater secondary	1.9	–	10	2.5
SAK-CH5-SS	Stormwater secondary	2.8	149	14 to 52.5	2.5

The stormwater main channels carry 154 to 795 m^3/s with bed widths of 10 to 267 m, and the stormwater secondary channels 25 to 336 m^3/s with bed widths of 10 to 72.5 m. The widest section is that of SHA-CH3-SM, 267 m at its outfall. The widths are those of the hydraulic sizing; the final section at each crossing and constraint, and the velocities along the steeper reaches, follow in the detailed design.

Figure 17 draws the bed of every reach to scale with its width; Figure 18 gives the 200-year design flow along the channels, rising from reach to reach as the tributaries join. The lengths by class are in Figure 19, the distribution of the designed widths, weighted by length, in Figure 20, and the full inventory of sizes in Table 9. Figure 21 relates the width of each reach to its design flow: at a given flow the width varies with the bed slope, the flatter reaches near the coast being the wider.

Designed bed width of the channels for the 200-year flow (m)

Flood Protection Scheme in North and South Al Batinah — Block 04

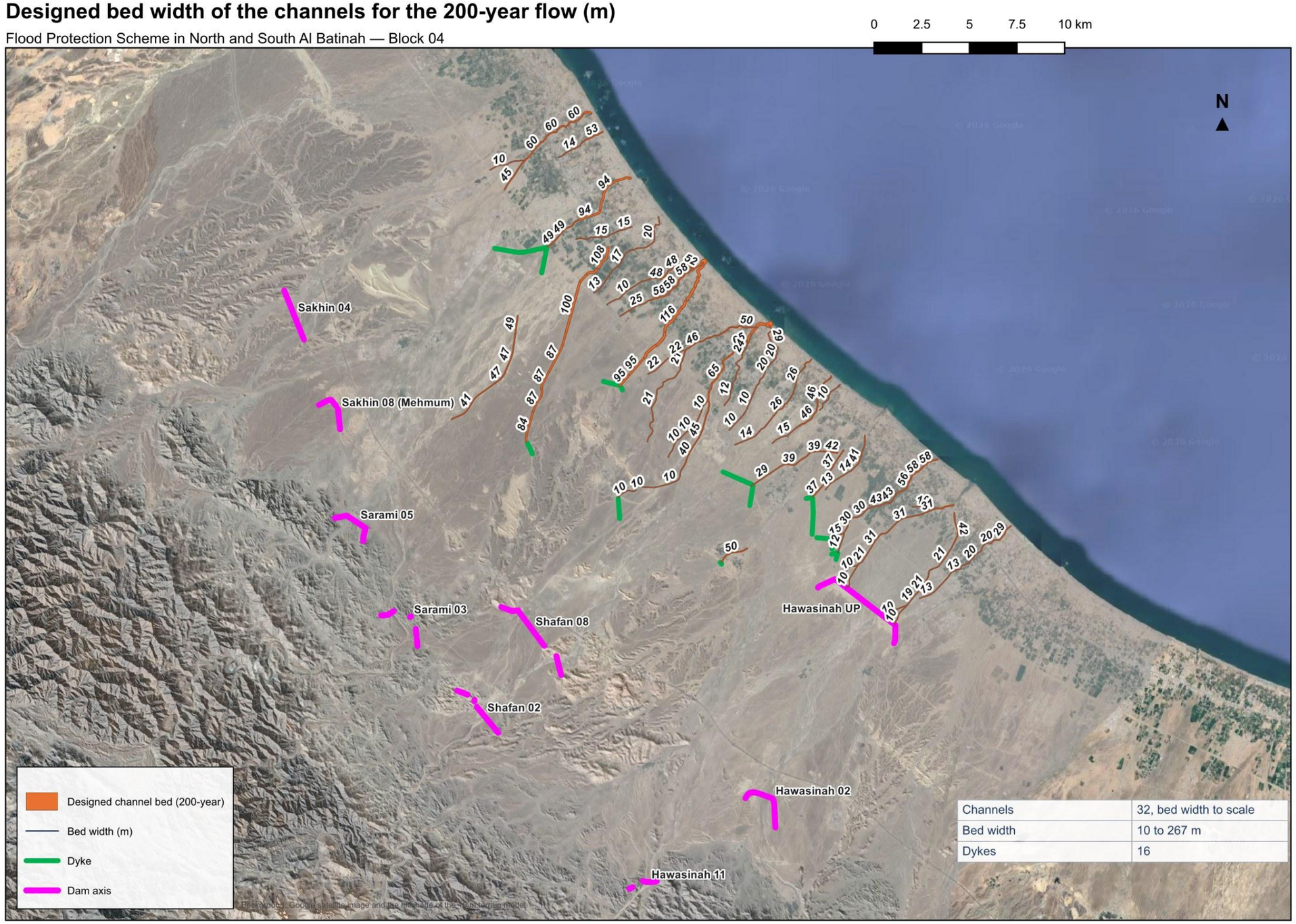


Figure 17 – Designed bed width of the channels for the 200-year flow (m)

200-year design flow along the channels (m³/s)

Flood Protection Scheme in North and South Al Batinah — Block 04

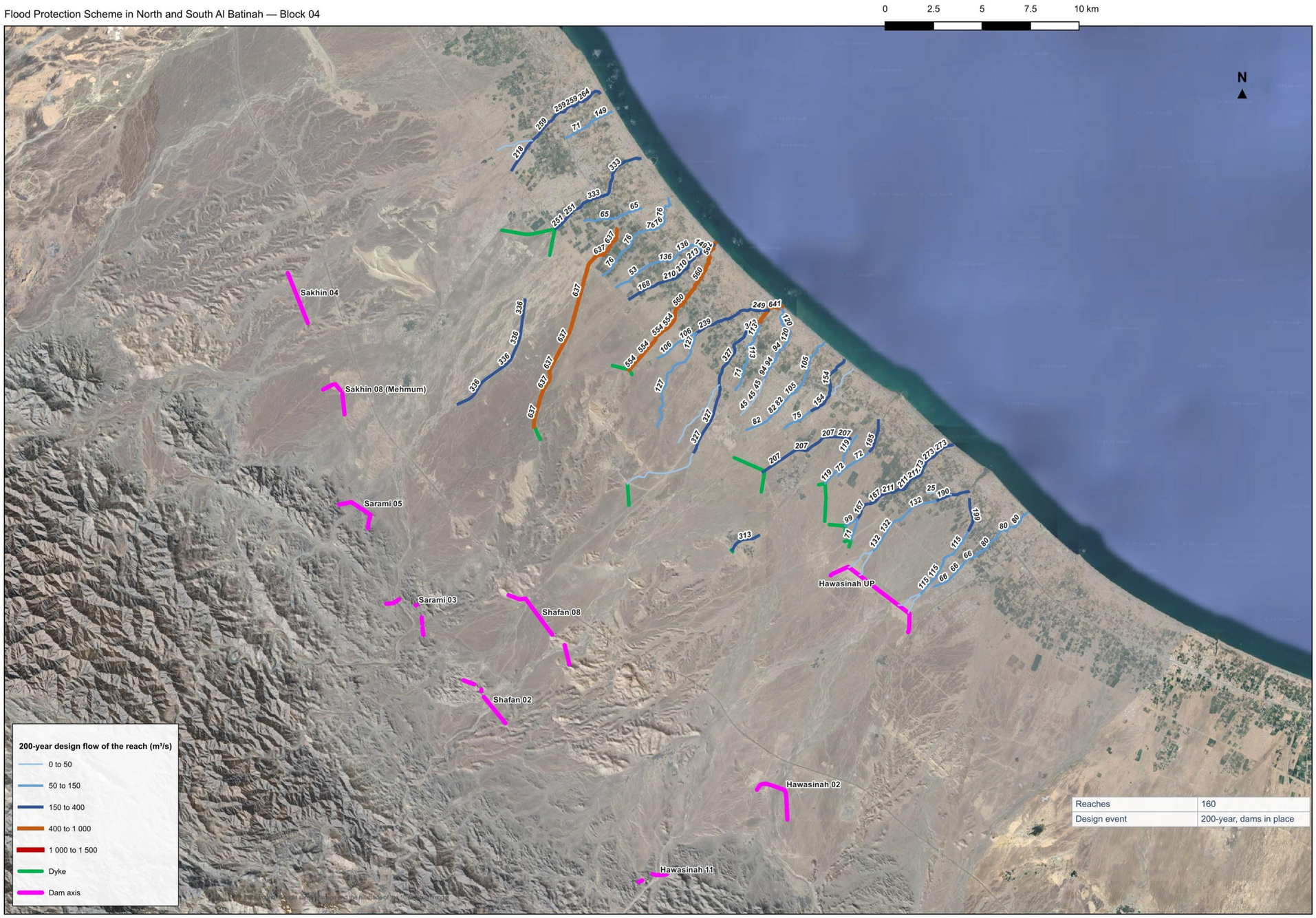


Figure 18 – 200-year design flow along the channels (m³/s)

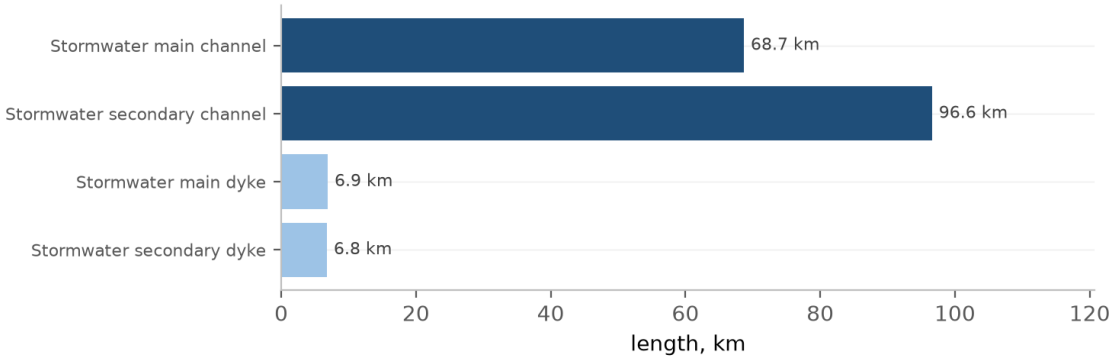


Figure 19 – Stormwater channels and dykes of Block 04 by class

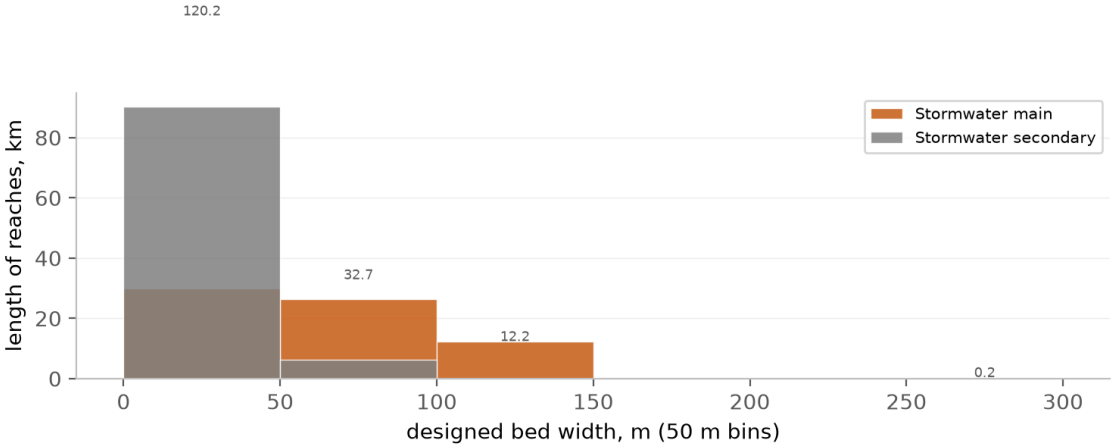


Figure 20 – Distribution of the designed bed widths, by length of reach, in 50 m bins

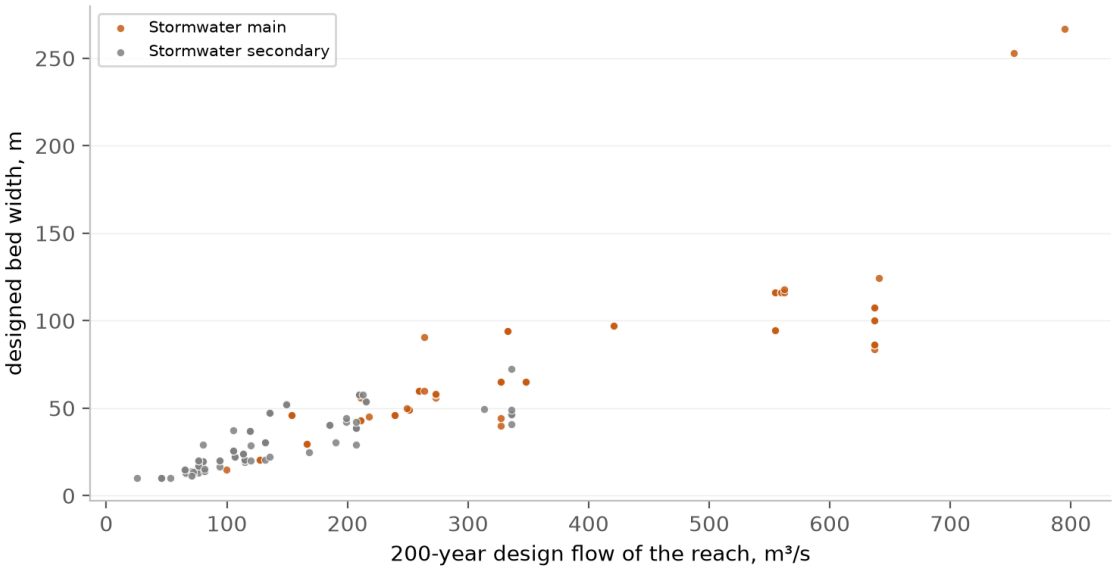


Figure 21 – Designed bed width against the 200-year design flow of the reach

Table 9 – Channel sizes inventory: depth x bed width (m) and the length of reaches of that size (m)

Size (m)	Length (m)	Size (m)	Length (m)	Size (m)	Length (m)
2.5 x 10	23,146	2.5 x 29	3,046	2.5 x 54	941
2.5 x 11.5	1,089	2.5 x 29.5	1,996	2.5 x 56	1,359
2.5 x 12	1,856	2.5 x 30.5	4,188	2.5 x 57.5	3,551
2.5 x 12.5	2,596	2.5 x 37	3,019	2.5 x 58	976
2.5 x 13	2,652	2.5 x 37.5	317	2.5 x 60	4,199
2.5 x 13.5	1,243	2.5 x 38.5	2,642	2.5 x 65	4,138
2.5 x 14	3,683	2.5 x 40	1,667	2.5 x 72.5	576
2.5 x 14.5	1,717	2.5 x 40.5	1,654	2.5 x 83.5	1,649
2.5 x 15	3,803	2.5 x 41	3,014	2.5 x 86.5	4,539
2.5 x 15.5	667	2.5 x 42	2,029	2.5 x 90.5	416
2.5 x 16.5	559	2.5 x 43	1,683	2.5 x 94	3,907
2.5 x 17	2,488	2.5 x 44.5	2,437	2.5 x 94.5	1,969
2.5 x 19	868	2.5 x 45	1,858	2.5 x 97	885
2.5 x 19.5	2,226	2.5 x 46	5,890	2.5 x 100	3,283
2.5 x 20	4,519	2.5 x 46.5	2,939	2.5 x 107.5	1,965
2.5 x 20.5	10,194	2.5 x 47.5	2,066	2.5 x 116	5,338
2.5 x 22	2,827	2.5 x 49	3,148	2.5 x 118	861
2.5 x 24	2,088	2.5 x 49.5	1,642	2.5 x 124.5	729
2.5 x 25	1,941	2.5 x 50	1,684	2.5 x 253	109
2.5 x 25.5	3,436	2.5 x 52	794	2.5 x 267	86
2.5 x 28.5	1,412	2.5 x 52.5	1,096		
Total	**165,298**				

The reaches are labelled channel by channel, catchment by catchment, in Figure 22 to Figure 25; their length, slope, velocity, size and flow are listed in Appendix A, and the plan and profile drawings in Appendix B. 61 of the 160 reaches run above 3 m/s at the 200-year flow on the steeper ground of the foothills; energy dissipation, bed protection or a flatter profile in steps will be set for them in the detailed design.

Channels and reach numbers: Al Hawasinah catchment (7 channels, 25 reaches)

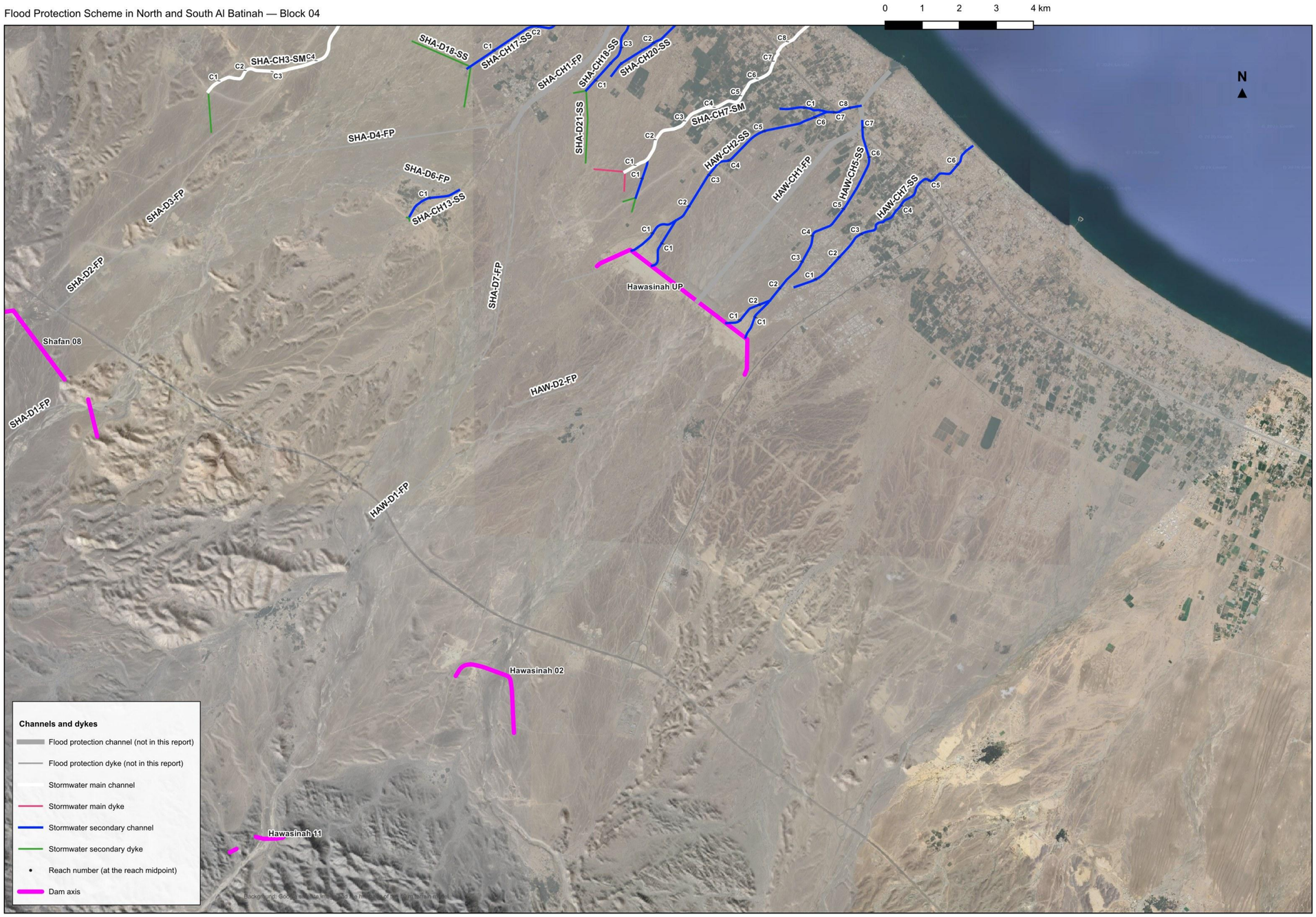


Figure 22 – Channels by class with their names and the reach numbers of the hydraulic model: Al Hawasinah catchment

Channels and reach numbers: Shafan catchment (21 channels, 113 reaches)

Flood Protection Scheme in North and South Al Batinah — Block 04

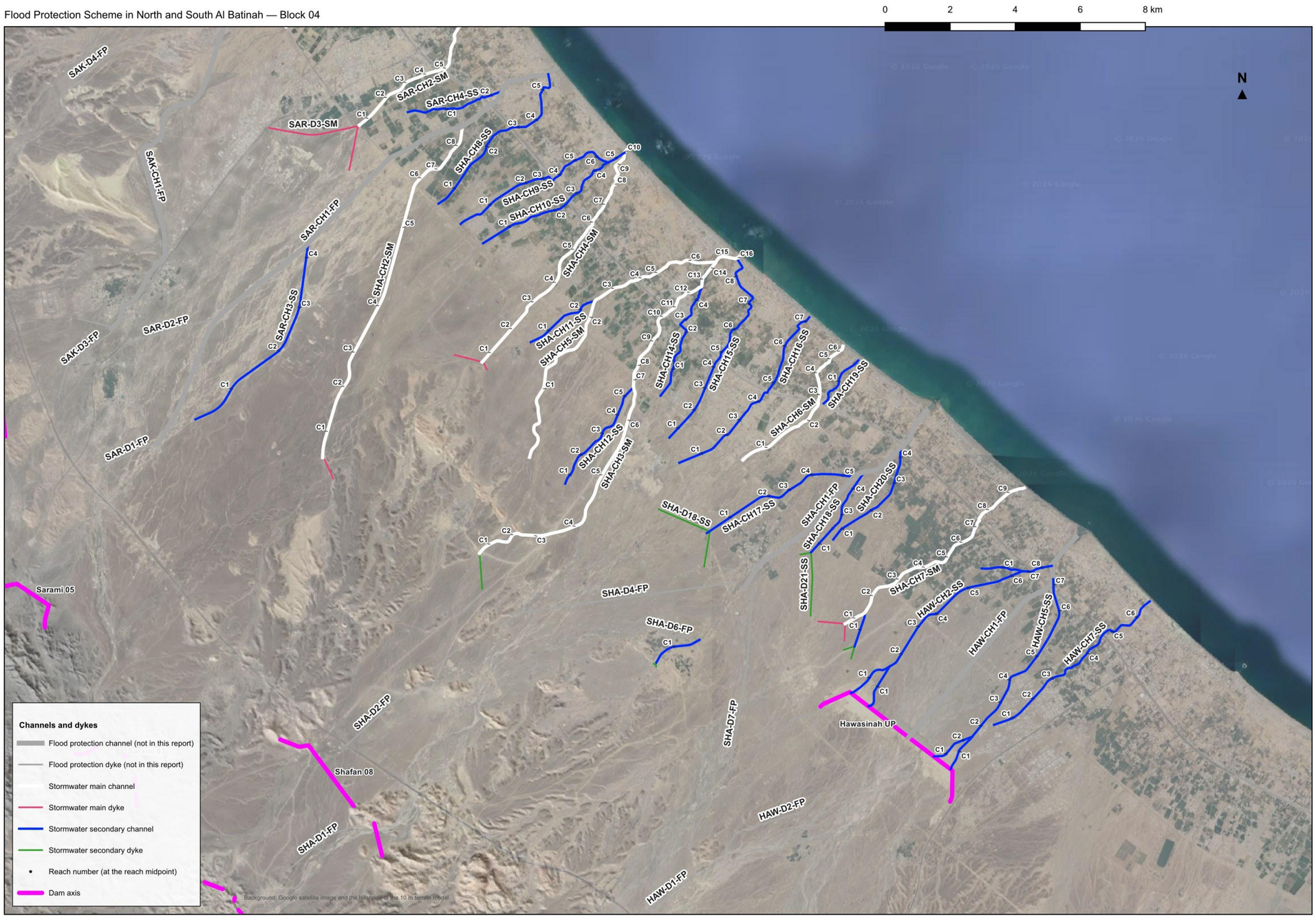


Figure 23 – Channels by class with their names and the reach numbers of the hydraulic model: Shafan catchment

Channels and reach numbers: Sarami catchment (4 channels, 12 reaches)

Flood Protection Scheme in North and South Al Batinah — Block 04

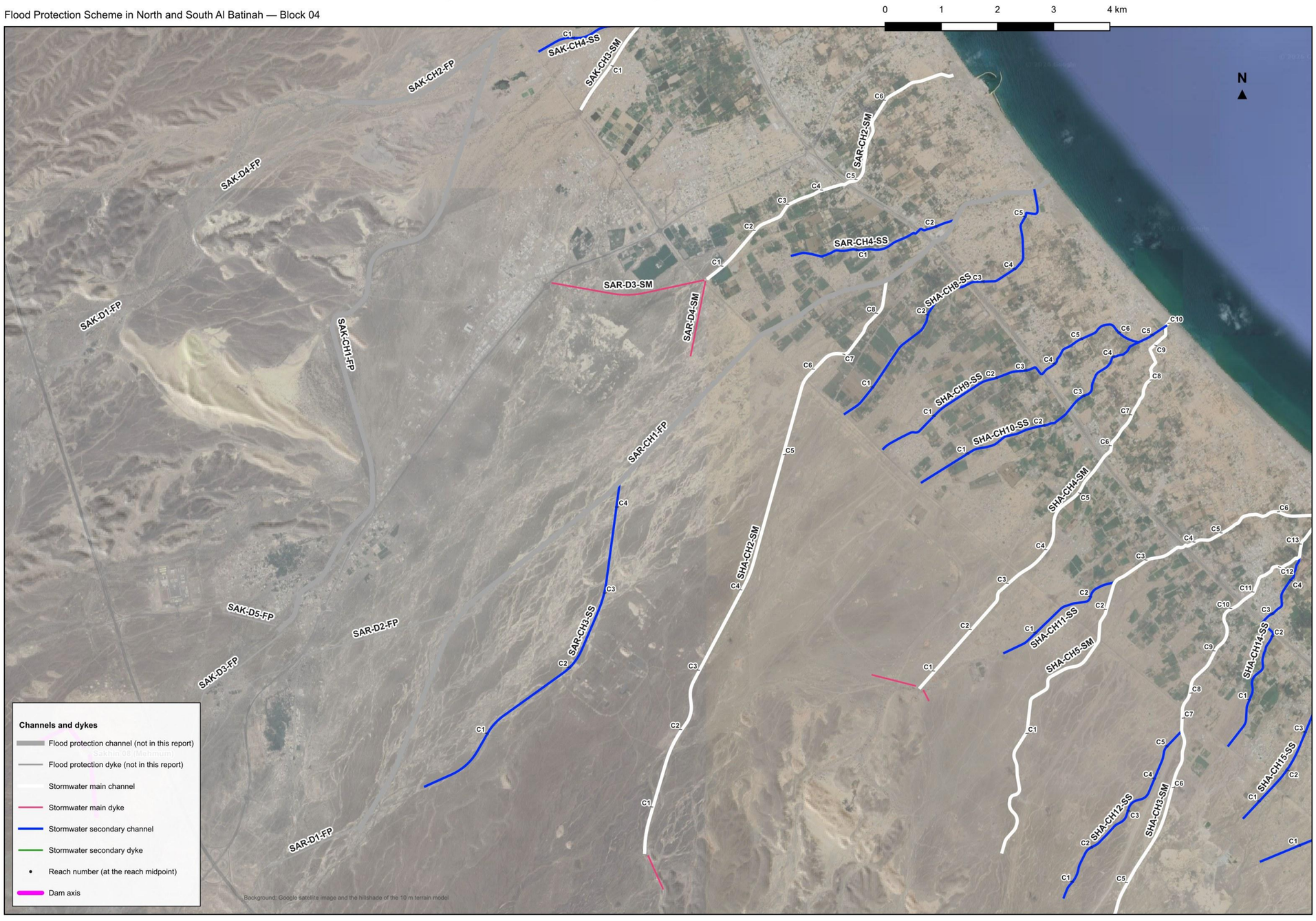


Figure 24 – Channels by class with their names and the reach numbers of the hydraulic model: Sarami catchment

Channels and reach numbers: Sakhin catchment (5 channels, 10 reaches)

Flood Protection Scheme in North and South Al Batinah — Block 04



Figure 25 – Channels by class with their names and the reach numbers of the hydraulic model: Sakhin catchment

7. OUTFALLS

Eleven of the 32 stormwater channels discharge directly to the sea, at the crossings of the coastal road; the others join a flood-protection channel or a main channel upstream of its outfall. At the 200-year flow the sea outfalls of the stormwater channels receive 0 to 795 m^3/s.

Figure 26 locates the outfalls with, at each, the designed invert level I, the existing terrain level Fix and their difference; the invert is printed in red where it lies below the existing terrain. Table 10 gives, for each outfall, the 200-year flow and bed width of its last reach, its invert and ground level, the flow depth and the invert against the mean higher high water of 0.70 m at Al Khaburah.

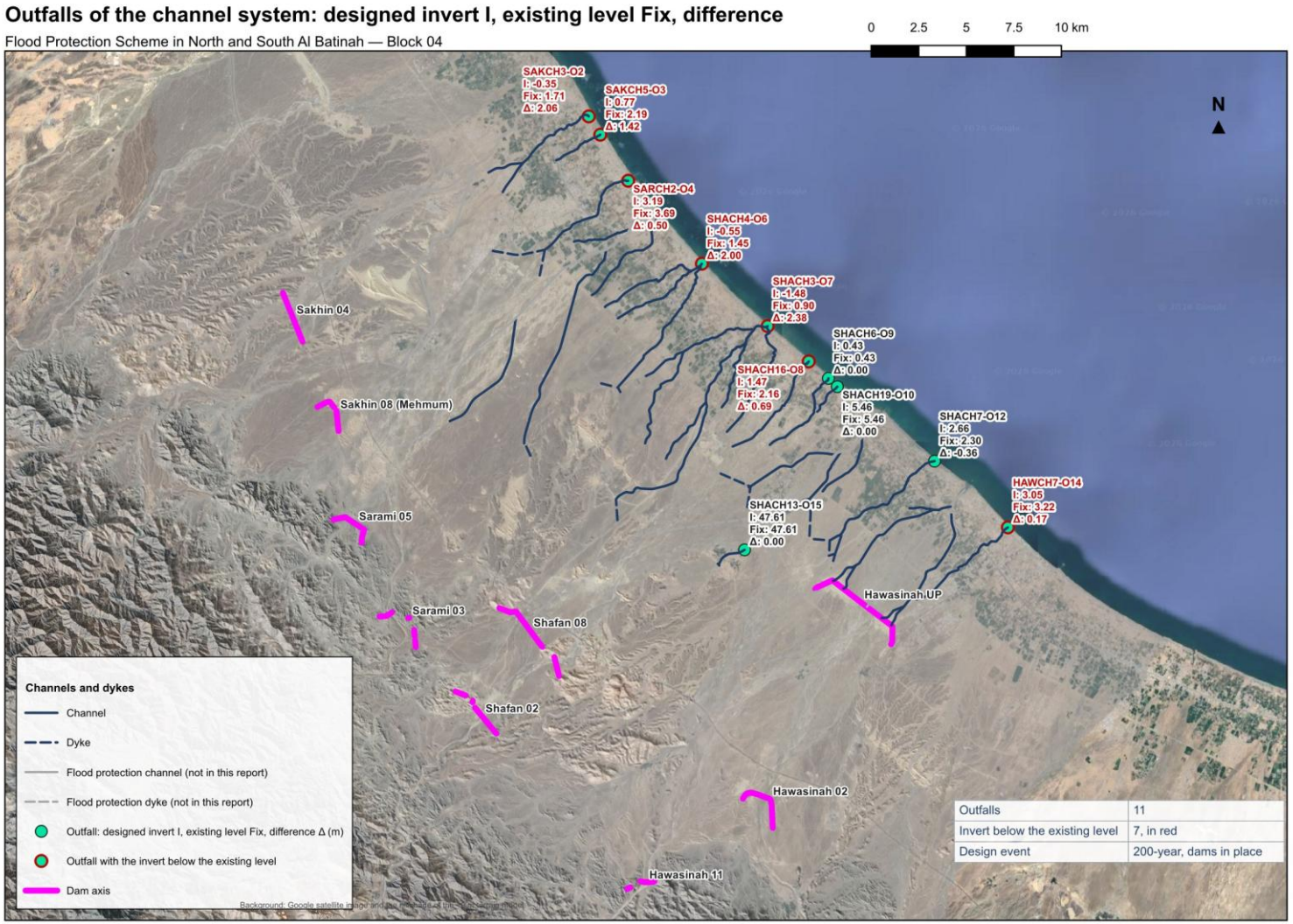


Figure 26 – Outfalls of the channel system: designed invert, existing terrain level and their difference (m)

Table 10 – Outfalls: 200-year flow, bed width, levels and tidal margin

Outfall	Channel	Q200 (m^3/s)	Bed width (m)	Invert (m)	Ground (m)	Flow depth (m)	Invert – MHHW (m)
HAWCH7-O14	HAW-CH7-SS	80	29	3.05	5.82	0.92	2.35
SAKCH3-O2	SAK-CH3-SM	264	90.5	-0.35	4.21	0.95	-1.05
SAKCH5-O3	SAK-CH5-SS	149	52.5	0.77	4.99	0.94	0.07
SARCH2-O4	SAR-CH2-SM	333	94	3.19	5.68	1.08	2.49
SHACH13-O15	SHA-CH13-SS	313	49.5	47.61	50.11	1.60	46.91
SHACH16-O8	SHA-CH16-SS	105	37.5	1.47	4.66	0.93	0.77
SHACH19-O10	SHA-CH19-SS	0	10	5.46	7.96	0.00	4.76
SHACH3-O7	SHA-CH3-SM	795	267	-1.48	3.40	0.97	-2.18
SHACH4-O6	SHA-CH4-SM	752	253	-0.55	3.95	0.97	-1.25
SHACH6-O9	SHA-CH6-SM	154	46	0.43	2.93	1.04	-0.27
SHACH7-O12	SHA-CH7-SM	273	58	2.66	5.16	1.31	1.96

8. CULVERTS AND BRIDGES

The channels cross the Batinah Highway at 17 points, the Batinah Coastal Road at 10 and the planned North Batinah Bypass at 22; the Batinah Expressway does not cross a Block 04 channel (Figure 27). Every crossing needs a structure that passes the designed section: a box culvert of the full bed width of the channel and the full channel depth as clear height, 2.5 m or 3 m, with the road embankment built above it. The bypass culverts are sized on its planned alignment and a corridor of 45 m.

Existing culverts stand at 9 of the crossings (Table 11); none has the width of the channel it would serve, and the other crossings have no structure. All are to be replaced or added.

Table 11 – Existing culverts at the crossings

ID	Road	Channel	Station (m)	Existing type	Cells x width x height (m)	Barrels
4	Batinah Highway	HAW-CH2-SS	7,120	–	–	–
5	North Batinah Bypass	HAW-CH2-SS	3,346	–	–	–
6	Batinah Highway	HAW-CH5-SS	6,770	–	–	–
7	North Batinah Bypass	HAW-CH5-SS	3,023	–	–	–
8	Batinah Coastal Road	HAW-CH7-SS	6,508	–	–	–
9	Batinah Highway	HAW-CH7-SS	4,562	–	–	–
10	North Batinah Bypass	HAW-CH7-SS	1,169	–	–	–
14	North Batinah Bypass	SHA-CH2-SM	8,933	–	–	–
15	Batinah Coastal Road	SHA-CH3-SM	13,878	–	–	–
16	Batinah Highway	SHA-CH3-SM	10,761	Culvert	2x2x1	1
17	North Batinah Bypass	SHA-CH3-SM	8,414	–	–	–
18	Batinah Coastal Road	SHA-CH4-SM	8,273	–	–	–
19	Batinah Highway	SHA-CH4-SM	5,334	–	–	–
20	North Batinah Bypass	SHA-CH4-SM	2,784	–	–	–
21	Batinah Highway	SHA-CH5-SM	7,208	Culvert	3x1.5x1	1
22	North Batinah Bypass	SHA-CH5-SM	4,748	–	–	–
23	Batinah Coastal Road	SHA-CH6-SM	5,159	–	–	–
24	Batinah Highway	SHA-CH6-SM	2,995	–	–	–
25	North Batinah Bypass	SHA-CH6-SM	0	–	–	–
26	Batinah Coastal Road	SHA-CH7-SM	7,395	–	–	–
27	Batinah Highway	SHA-CH7-SM	5,708	Culvert	1x2x1	1
28	North Batinah Bypass	SHA-CH7-SM	1,851	–	–	–
29	Batinah Highway	SHA-CH8-SS	3,326	–	–	–
30	North Batinah Bypass	SHA-CH8-SS	0	–	–	–
31	Batinah Highway	SHA-CH9-SS	3,355	–	–	–
32	North Batinah Bypass	SHA-CH9-SS	0	–	–	–
33	Batinah Highway	SHA-CH10-SS	2,974	Culvert	1x2x1	1
34	North Batinah Bypass	SHA-CH10-SS	0	–	–	–
35	North Batinah Bypass	SHA-CH11-SS	1,350	–	–	–
36	Batinah Highway	SHA-CH14-SS	2,539	Culvert	2x2x1	1
37	North Batinah Bypass	SHA-CH14-SS	0	–	–	–
38	Batinah Highway	SHA-CH15-SS	2,910	Culvert	2x2x1	1

ID	Road	Channel	Station (m)	Existing type	Cells x width x height (m)	Barrels
39	North Batinah Bypass	SHA-CH15-SS	839	–	–	–
40	Batinah Coastal Road	SHA-CH16-SS	6,414	–	–	–
41	Batinah Highway	SHA-CH16-SS	3,634	Culvert	2x2x1	1
42	North Batinah Bypass	SHA-CH16-SS	1,397	–	–	–
43	North Batinah Bypass	SHA-CH17-SS	2,686	–	–	–
44	North Batinah Bypass	SHA-CH18-SS	733	–	–	–
45	Batinah Coastal Road	SHA-CH19-SS	1,807	–	–	–
46	North Batinah Bypass	SHA-CH20-SS	0	–	–	–
50	Batinah Coastal Road	SAR-CH2-SM	6,132	–	–	–
51	Batinah Highway	SAR-CH2-SM	3,086	Culvert	3x2x1	1
52	North Batinah Bypass	SAR-CH2-SM	0	–	–	–
53	Batinah Highway	SAR-CH4-SS	2,398	Culvert	1x2x1	1
57	Batinah Coastal Road	SAK-CH3-SM	6,473	–	–	–
58	Batinah Highway	SAK-CH3-SM	3,693	–	–	–
59	North Batinah Bypass	SAK-CH3-SM	0	–	–	–
60	North Batinah Bypass	SAK-CH4-SS	0	–	–	–
61	Batinah Coastal Road	SAK-CH5-SS	2,774	–	–	–

The proposed culverts are in Table 12: box cells of 4.0 m clear width from the standard combinations of the Oman Highway Design Standard, the number of cells from the bed width, the barrel on the slope of the channel it carries, and the headwater checked with the inlet- and outlet-control equations of the FHWA culvert design method (square-edged inlet with wingwalls, entrance loss 0.4, barrel roughness 0.015, the tailwater at the normal depth of the channel downstream): every structure keeps its headwater below 1.2 times the clear height. The length across the road is the road corridor, 45 m on the Batinah Highway and the bypass and 32 m on the Coastal Road.

Table 12 – Proposed box culverts at the road crossings (200-year flow)

ID	Road	Channel	Q200 (m^3/s)	Channel W x D (m)	Cells x W x H (m)	Overall width (m)	Length (m)	Slope (m/m)	HW / H	Cost (million OMR)
4	Highway	HAW-CH2-SS	215	54 x 2.5	14 x 4 x 2.5	59.8	45	0.0019	0.75	0.36
5	Bypass	HAW-CH2-SS	132	30.5 x 2.5	8 x 4 x 2.5	34.2	45	0.0024	0.75	0.21
6	Highway	HAW-CH5-SS	199	44.5 x 2.5	12 x 4 x 2.5	51.2	45	0.0036	0.75	0.31
7	Bypass	HAW-CH5-SS	115	20.5 x 2.5	6 x 4 x 2.5	25.8	45	0.0045	0.82	0.16
8	Coastal Road	HAW-CH7-SS	80	29 x 2.5	8 x 4 x 2.5	34.2	32	0.0010	0.75	0.15
9	Highway	HAW-CH7-SS	80	19.5 x 2.5	5 x 4 x 2.5	21.5	45	0.0070	0.73	0.13
10	Bypass	HAW-CH7-SS	66	12.5 x 2.5	4 x 4 x 2.5	17.2	45	0.0045	0.75	0.11
14	Bypass	SHA-CH2-SM	637	100 x 2.5	25 x 4 x 2.5	106.5	45	0.0047	1.00	0.64
15	Coastal Road	SHA-CH3-SM	795	267 x 2.5	67 x 4 x 2.5	285	32	0.0010	0.75	1.22
16	Highway	SHA-CH3-SM	348	65 x 2.5	17 x 4 x 2.5	72.5	45	0.0041	0.86	0.48
17	Bypass	SHA-CH3-SM	327	65 x 2.5	17 x 4 x 2.5	72.5	45	0.0031	0.83	0.44
18	Coastal Road	SHA-CH4-SM	752	253 x 2.5	64 x 4 x 2.5	272.2	32	0.0010	0.75	1.16
19	Highway	SHA-CH4-SM	560	116 x 2.5	29 x 4 x 2.5	123.5	45	0.0037	0.83	0.74
20	Bypass	SHA-CH4-SM	554	116 x 2.5	29 x 4 x 2.5	123.5	45	0.0026	0.83	0.74
21	Highway	SHA-CH5-SM	239	46 x 2.5	12 x 4 x 2.5	51.2	45	0.0066	0.85	0.34
22	Bypass	SHA-CH5-SM	127	20.5 x 2.5	6 x 4 x 2.5	25.8	45	0.0055	0.88	0.16

ID	Road	Channel	Q200 (m^3/s)	Channel W x D (m)	Cells x W x H (m)	Overall width (m)	Length (m)	Slope (m/m)	HW / H	Cost (million OMR)
23	Coastal Road	SHA-CH6-SM	154	46 x 2.5	12 x 4 x 2.5	51.2	32	0.0082	0.63	0.22
24	Highway	SHA-CH6-SM	154	46 x 2.5	12 x 4 x 2.5	51.2	45	0.0014	0.75	0.31
25	Bypass	SHA-CH6-SM	75	14.5 x 2.5	4 x 4 x 2.5	17.2	45	0.0040	0.81	0.11
26	Coastal Road	SHA-CH7-SM	273	58 x 2.5	15 x 4 x 2.5	64	32	0.0027	0.80	0.28
27	Highway	SHA-CH7-SM	273	56 x 2.5	14 x 4 x 2.5	59.8	45	0.0039	0.84	0.40
28	Bypass	SHA-CH7-SM	166	29.5 x 2.5	8 x 4 x 2.5	34.2	45	0.0042	0.87	0.21
29	Highway	SHA-CH8-SS	76	17 x 2.5	5 x 4 x 2.5	21.5	45	0.0062	0.71	0.13
30	Bypass	SHA-CH8-SS	76	13 x 2.5	4 x 4 x 2.5	17.2	45	0.0058	0.82	0.11
31	Highway	SHA-CH9-SS	136	47.5 x 2.5	12 x 4 x 2.5	51.2	45	0.0071	0.58	0.31
32	Bypass	SHA-CH9-SS	53	10 x 2.5	3 x 4 x 2.5	13	45	0.0055	0.78	0.08
33	Highway	SHA-CH10-SS	210	57.5 x 2.5	15 x 4 x 2.5	64	45	0.0084	0.66	0.42
34	Bypass	SHA-CH10-SS	168	25 x 2.5	7 x 4 x 2.5	30	45	0.0061	0.96	0.18
35	Bypass	SHA-CH11-SS	106	22 x 2.5	6 x 4 x 2.5	25.8	45	0.0033	0.78	0.16
36	Highway	SHA-CH14-SS	113	24 x 2.5	6 x 4 x 2.5	25.8	45	0.0046	0.82	0.17
37	Bypass	SHA-CH14-SS	71	12 x 2.5	3 x 4 x 2.5	13	45	0.0061	0.95	0.08
38	Highway	SHA-CH15-SS	94	20 x 2.5	5 x 4 x 2.5	21.5	45	0.0031	0.82	0.15
39	Bypass	SHA-CH15-SS	45	10 x 2.5	3 x 4 x 2.5	13	45	0.0118	0.70	0.08
40	Coastal Road	SHA-CH16-SS	105	37.5 x 2.5	10 x 4 x 2.5	42.8	32	0.0010	0.75	0.19
41	Highway	SHA-CH16-SS	105	25.5 x 2.5	7 x 4 x 2.5	30	45	0.0023	0.73	0.20
42	Bypass	SHA-CH16-SS	82	14 x 2.5	4 x 4 x 2.5	17.2	45	0.0053	0.86	0.11
43	Bypass	SHA-CH17-SS	207	38.5 x 2.5	10 x 4 x 2.5	42.8	45	0.0037	0.87	0.26
44	Bypass	SHA-CH18-SS	119	37 x 2.5	10 x 4 x 2.5	42.8	45	0.0116	0.59	0.26
45	Coastal Road	SHA-CH19-SS	–	10 x 2.5	3 x 4 x 2.5	13	32	0.0024	–	0.06
46	Bypass	SHA-CH20-SS	72	12.5 x 2.5	4 x 4 x 2.5	17.2	45	0.0056	0.79	0.11
50	Coastal Road	SAR-CH2-SM	333	94 x 2.5	24 x 4 x 2.5	102.2	32	0.0031	0.70	0.44
51	Highway	SAR-CH2-SM	333	94 x 2.5	24 x 4 x 2.5	102.2	45	0.0067	0.66	0.67
52	Bypass	SAR-CH2-SM	251	49 x 2.5	13 x 4 x 2.5	55.5	45	0.0033	0.83	0.34
53	Highway	SAR-CH4-SS	65	15 x 2.5	4 x 4 x 2.5	17.2	45	0.0087	0.74	0.12
57	Coastal Road	SAK-CH3-SM	264	90.5 x 2.5	23 x 4 x 2.5	98	32	0.0010	0.75	0.42
58	Highway	SAK-CH3-SM	259	60 x 2.5	15 x 4 x 2.5	64	45	0.0023	0.77	0.39
59	Bypass	SAK-CH3-SM	218	45 x 2.5	12 x 4 x 2.5	51.2	45	0.0029	0.80	0.31
60	Bypass	SAK-CH4-SS	–	10 x 2.5	3 x 4 x 2.5	13	45	0.0039	–	0.08
61	Coastal Road	SAK-CH5-SS	149	52.5 x 2.5	14 x 4 x 2.5	59.8	32	0.0010	0.75	0.26
	Total				**637 cells**					**14.95**

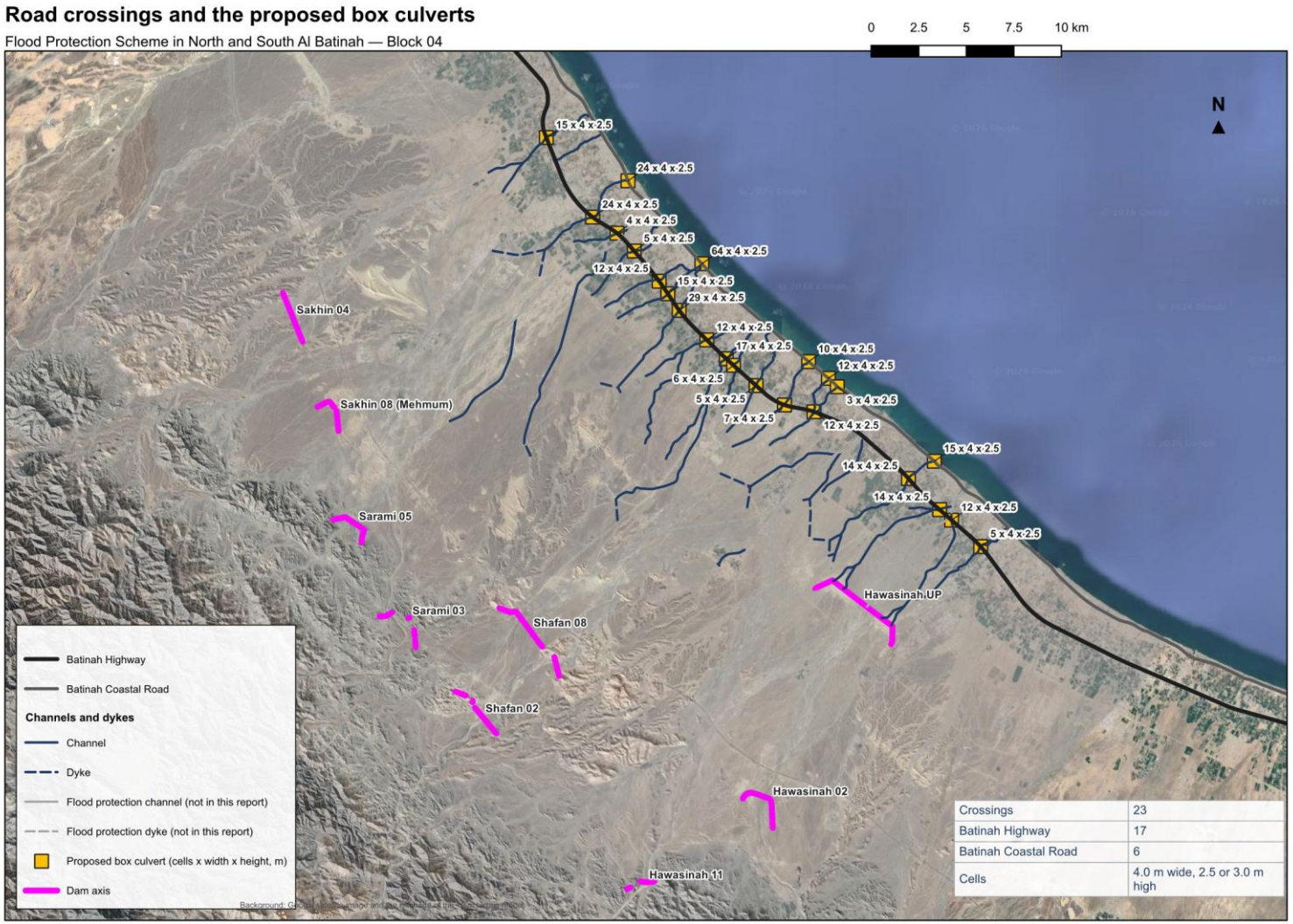


Figure 27 – Road crossings and the proposed box culverts

9. DYKES

The dykes keep the flood from spreading before it reaches the channels and lead it to their entrances. The 16 dykes, 13.7 km in all, stand 1.6 to 3.6 m high (Table 13, Figure 28); the highest, SHA-D17-SS, closes a ponding area where the 200-year depth reaches 3.5 m.

Table 13 – Dykes: 200-year depth along the line, adopted height and cost

ID	Dyke	Class	Length (m)	Depth, mean (m)	Depth, 90 % (m)	Depth, max (m)	Height (m)	Cost (million OMR)
13	SHA-D10-SM	Stormwater main	638	1.99	2.69	3.51	3.1	0.13
14	SHA-D11-SM	Stormwater main	787	0.82	1.80	2.14	2.0	0.10
15	SHA-D12-SM	Stormwater main	194	0.94	2.00	2.31	2.1	0.03
16	SHA-D13-SM	Stormwater main	758	0.78	1.16	1.69	1.9	0.09
17	SHA-D14-SM	Stormwater main	470	0.62	0.93	1.38	1.8	0.05
18	SHA-D15-SS	Stormwater secondary	1,058	0.44	0.74	1.16	1.6	0.11
19	SHA-D16-SS	Stormwater secondary	47	1.77	3.09	3.46	2.9	0.01
20	SHA-D17-SS	Stormwater secondary	72	2.49	3.27	3.49	3.6	0.02
21	SHA-D18-SS	Stormwater secondary	1,668	0.59	0.94	1.75	1.7	0.18
22	SHA-D19-SS	Stormwater secondary	1,069	0.76	1.15	1.53	1.9	0.13
23	SHA-D20-SS	Stormwater secondary	309	1.25	1.51	1.76	2.4	0.05
24	SHA-D21-SS	Stormwater secondary	1,920	0.60	1.04	1.77	1.8	0.22
25	SHA-D22-SS	Stormwater secondary	300	0.94	1.21	1.82	2.1	0.04
26	SHA-D23-SS	Stormwater secondary	353	1.02	1.50	1.93	2.2	0.05
29	SAR-D3-SM	Stormwater main	2,734	1.01	1.88	2.45	2.2	0.38
30	SAR-D4-SM	Stormwater main	1,337	0.99	1.42	2.06	2.1	0.18
	Total		**13,715**					**1.75**

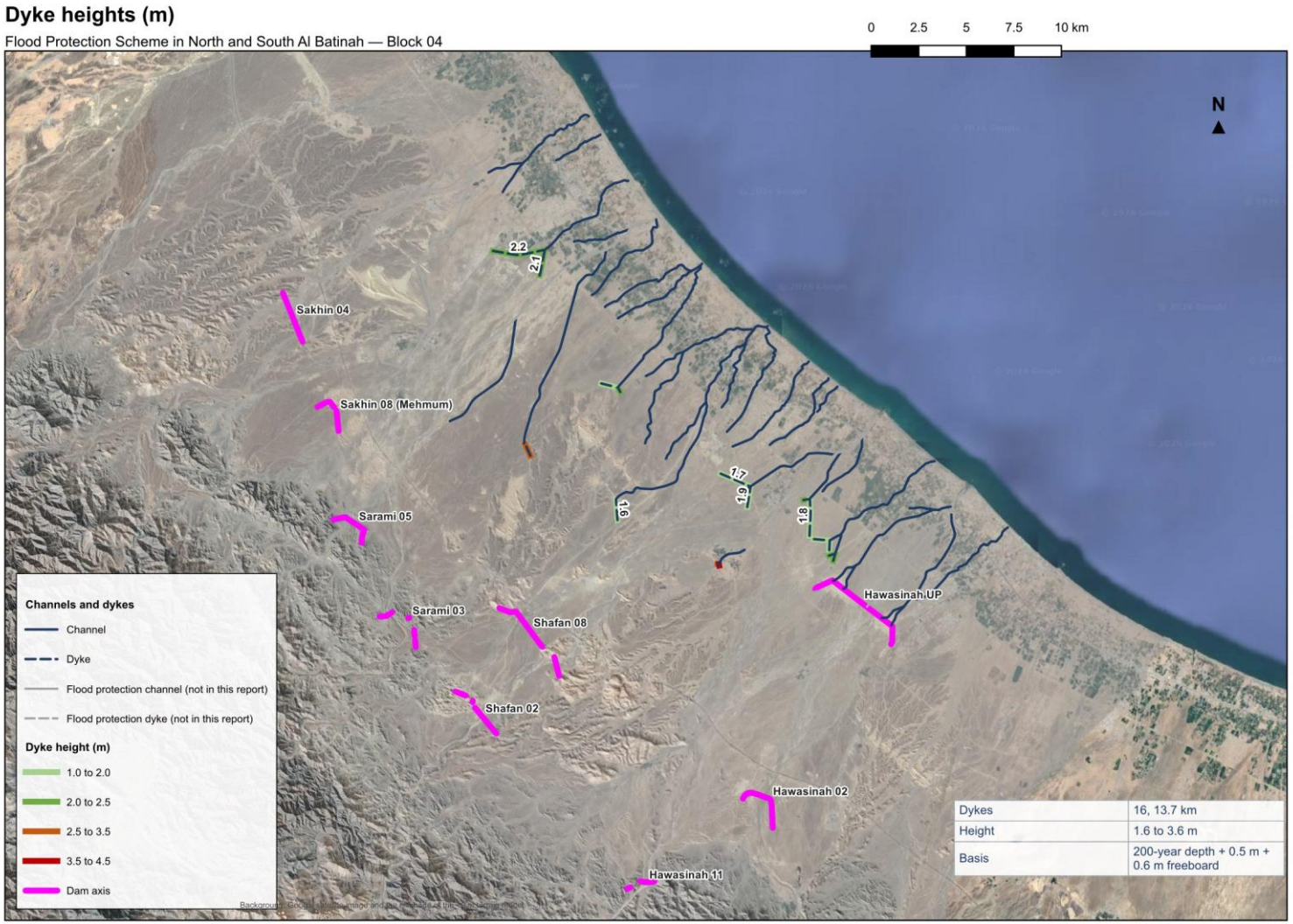


Figure 28 – Dyke heights (m)

10. QUANTITIES AND COST ESTIMATE

The quantities of the channels are measured on the designed sections against the 0.5 m terrain model of the project, reach by reach: the channel surface of the typical sections is built along each reach at its bed width, depth and invert, and the excavation and the fill are the volumes between that surface and the ground. The riprap is the two side slopes of every reach, 0.3 m thick as priced; the clearing is the footprint of the works. The dykes follow the typical fill section at their height, with a 0.4 m strip under the footprint. The culverts are priced by their concrete and reinforcement, blinding, waterproofing, the road works over them and the removal of the existing structure. The unit rates are those of the master plan estimate (Table 14).

Table 14 – Unit rates of the estimate

Item	Unit	Rate (OMR)
Clearing and grubbing	m^2	0.30
Excavation	m^3	2.10
Embankment and dyke fill	m^3	4.00
Riprap, 0.3 m, over geotextile	m^3	13.20
Reinforced concrete of the culverts	m^3	106.90
Reinforcement, 120 kg/m^3	t	417.10
Blinding concrete	m^3	45.00
Pavement and road works over the culverts	m^2	per the master plan build-up

Table 15 – Channels: earthworks, protection, land and cost

Channel	Length (km)	Cut (m^3)	Fill (m^3)	Clearing (m^2)	Riprap (m^2)	Corridor (ha)	In cadastral parcels (ha)	Construction (million OMR)	Land (million OMR)	Channel cost (million OMR)
HAW-CH2-SS	7.7	435,720	59,850	336,312	69,611	45.6	10.7	1.53	5.64	7.17
HAW-CH3-SS	1.4	44,099	3,037	35,076	12,985	5.9	3.2	0.17	2.39	2.55
HAW-CH4-SS	1.4	45,553	4,069	32,264	12,488	5.7	1.4	0.17	1.07	1.24
HAW-CH5-SS	7.1	370,292	56,291	277,740	63,699	38.6	11.9	1.34	3.66	5.00
HAW-CH6-SS	1.3	42,278	2,759	31,640	11,451	5.2	0.3	0.15	0.17	0.33
HAW-CH7-SS	6.5	209,914	95,544	240,484	58,666	32.0	2.4	1.13	0.34	1.46
SHA-CH2-SM	11.4	2,305,908	75,612	1,245,032	103,096	142.6	28.2	5.93	1.49	7.42
SHA-CH3-SM	13.9	1,892,850	67,075	887,264	125,114	110.2	41.6	5.00	18.12	23.13
SHA-CH4-SM	8.3	1,893,135	105,785	1,071,788	74,616	119.0	49.4	5.02	18.43	23.44
SHA-CH5-SM	9.9	543,058	225,722	497,652	89,399	62.2	16.8	2.55	2.86	5.40
SHA-CH6-SM	5.2	480,366	31,311	260,536	46,509	34.3	9.0	1.40	1.75	3.15
SHA-CH7-SM	7.4	826,218	38,527	428,176	66,764	54.4	18.0	2.28	3.34	5.62
SHA-CH8-SS	5.9	230,779	37,332	189,292	53,050	28.3	2.2	0.90	1.14	2.04
SHA-CH9-SS	5.5	482,712	13,434	246,408	49,485	34.1	18.3	1.34	7.98	9.32
SHA-CH10-SS	5.4	687,369	28,644	340,048	48,447	42.1	23.2	1.85	3.93	5.78
SHA-CH11-SS	2.4	65,857	40,055	98,972	21,884	12.8	2.6	0.41	1.09	1.50
SHA-CH12-SS	3.8	97,385	16,881	94,724	34,016	15.5	3.6	0.43	2.14	2.58
SHA-CH13-SS	1.6	43,438	37,438	112,704	14,804	13.2	0.8	0.33	0.47	0.80
SHA-CH14-SS	3.9	224,150	7,405	127,808	35,561	19.5	4.2	0.68	0.85	1.52
SHA-CH15-SS	6.8	420,156	68,617	226,048	61,372	33.4	10.0	1.47	3.12	4.59

Channel	Length (km)	Cut (m^3)	Fill (m^3)	Clearing (m^2)	Riprap (m^2)	Corridor (ha)	In cadastral parcels (ha)	Construction (million OMR)	Land (million OMR)	Channel cost (million OMR)
SHA-CH16-SS	6.4	391,780	43,973	239,308	57,926	33.7	16.5	1.30	6.33	7.63
SHA-CH17-SS	5.1	417,888	20,764	256,864	46,053	34.1	1.0	1.22	0.29	1.51
SHA-CH18-SS	3.0	244,545	22,809	156,160	27,220	20.5	1.3	0.76	0.92	1.68
SHA-CH19-SS	1.8	39,463	22,281	49,424	16,335	7.4	3.6	0.25	0.54	0.79
SHA-CH20-SS	3.8	253,599	8,819	148,412	33,855	21.1	1.1	0.75	0.09	0.84
SHA-CH21-SS	1.1	39,366	3,026	26,852	9,817	4.6	0.7	0.14	0.45	0.59
SAR-CH2-SM	6.1	1,002,187	131,147	584,980	55,373	66.7	28.6	3.02	7.11	10.14
SAR-CH3-SS	6.9	801,510	12,437	396,988	61,901	51.9	0.2	2.10	0.14	2.24
SAR-CH4-SS	3.1	63,983	46,321	102,312	27,987	14.3	1.7	0.46	0.53	0.99
SAK-CH3-SM	6.5	1,122,683	12,105	468,076	58,354	57.4	17.2	2.78	2.96	5.73
SAK-CH4-SS	1.9	60,593	5,082	46,480	17,320	7.9	0.0	0.23	0.00	0.23
SAK-CH5-SS	2.8	275,133	5,950	123,864	25,004	16.7	10.9	0.74	4.35	5.09
Total	**165.3**	**16,053,967**	**1,350,102**	**9,379,688**	**1,490,162**	**1,191.0**		**47.83**	**103.69**	**151.52**

The corridors of the channels, the bed with the embankments and an access road on each side, take 1,191 ha, of which 341 ha lie in the parcels of the cadastre (Table 16). The land is priced at the rates of the master plan's estimate of expropriation: agricultural land 50 000 OMR/ha, residential 755 000 OMR/ha, commercial 1.49 million OMR/ha and industrial 1.5 million OMR/ha; government, planned and vacant land carries no cost. The land of the corridors comes to 103.7 million OMR.

Table 16 – Land take of the channel corridors by land use of the cadastre and its cost

Land use	Area (ha)	Rate (OMR/ha)	Cost (million OMR)
Agricultural	196.6	50,189	9.87
Residential	122.8	754,797	92.72
Government	17.2	0	0.00
Not Classified	2.8	0	0.00
Residential Commercial	1.0	754,797	0.78
Commercial	0.2	1,486,900	0.33
Total	**340.7**		**103.69**

Table 17 – Cost estimate of the stormwater channels and dykes of Block 04 (million OMR)

Item	Quantity	Cost (million OMR)
Channels, construction	165.3 km, 32 channels	47.83
Channels, land of the corridors	1,191 ha	103.69
Dykes	13.7 km, 16 dykes	1.75
Culverts	49 structures, 637 cells	14.95
Total		**168.22**

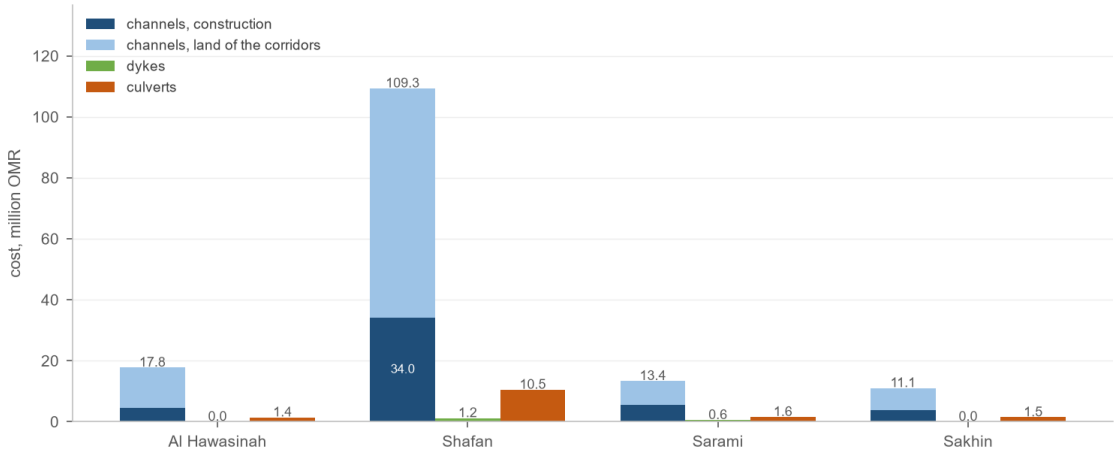


Figure 29 – Cost of the channels (construction and land), dykes and culverts by catchment

11. CONCLUSION

The 32 stormwater channels of Block 04 drain the plain between the flood-protection channels to the sea. Their alignments follow the wadis within the constraints of the Batinah Expressway, the Batinah Highway, the coastal road, the planned bypass, the OQ pipelines and the cadastral land use. They are designed for the 200-year runoff of their catchments with the dams in place, with the criteria of Section 6; their sections (Table 8) carry the 200-year flow with bed widths of 10 to 267 m: the main channels 154 to 795 m^3/s, the secondary channels 25 to 336 m^3/s.

At the 49 crossings of the Batinah Highway, the Coastal Road and the planned bypass box culverts of the full bed width and the full channel depth are proposed, 637 cells of 4 m in all, checked for headwater. The 16 stormwater dykes, 13.7 km, are set at 1.6 to 3.6 m on the 200-year flood depth of the plain with a margin and the freeboard; the two-dimensional simulation of the scheme will refine them.

The estimate of the stormwater works comes to 168.2 million OMR at the unit rates and land rates of the master plan: 47.8 million for the construction of the channels (16.1 million m^3 of excavation, 1.4 million m^3 of fill), 103.7 million for the land of their corridors, 1.8 million for the dykes and 14.9 million for the culverts. The long-term effectiveness of the scheme depends on the reservation of the wadi corridors against development, the removal of the encroachments that narrow them, and the maintenance of the channels and culverts after every flood.

APPENDIX A – REACH CHARACTERISTICS AT THE 200-YEAR FLOW

Every reach of the hydraulic model with its length, bed slope, roughness, velocity, size (depth x bed width) and 200-year design flow. Reaches without a modelled inflow carry no flow and take the minimum section.

Label	Length (m)	Slope (m/m)	Manning's n	Velocity (m/s)	Size (m)	Flow (m^3/s)
HAWCH2-C1	1,508	0.0066	0.030	–	2.5 x 10	–
HAWCH2-C2	1,085	0.0059	0.030	3.47	2.5 x 20.5	132
HAWCH2-C3	902	0.0024	0.030	2.31	2.5 x 30.5	132
HAWCH2-C4	500	0.0100	0.030	3.59	2.5 x 30.5	132
HAWCH2-C5	2,046	0.0038	0.030	2.66	2.5 x 30.5	132
HAWCH2-C6	740	0.0052	0.030	3.37	2.5 x 30.5	190
HAWCH2-C7	364	0.0019	0.030	2.13	2.5 x 54	215
HAWCH2-C8	578	0.0035	0.030	2.55	2.5 x 54	215
HAWCH3-C1	1,440	0.0082	0.030	–	2.5 x 10	–
HAWCH4-C1	801	0.0079	0.030	–	2.5 x 10	–
HAWCH4-C2	584	0.0059	0.030	–	2.5 x 10	–
HAWCH5-C1	1,214	0.0068	0.030	–	2.5 x 10	–
HAWCH5-C2	868	0.0053	0.030	3.25	2.5 x 19	115
HAWCH5-C3	1,012	0.0045	0.030	3.03	2.5 x 20.5	115
HAWCH5-C4	500	0.0100	0.030	3.89	2.5 x 20.5	115
HAWCH5-C5	1,813	0.0060	0.030	3.31	2.5 x 20.5	115
HAWCH5-C6	1,325	0.0028	0.030	2.54	2.5 x 42	199
HAWCH5-C7	333	0.0036	0.030	2.69	2.5 x 44.5	199
HAWCH6-C1	1,270	0.0034	0.030	1.98	2.5 x 10	25
HAWCH7-C1	1,238	0.0045	0.030	2.83	2.5 x 12.5	66
HAWCH7-C2	500	0.0110	0.030	3.81	2.5 x 12.5	66
HAWCH7-C3	1,260	0.0040	0.030	2.71	2.5 x 13	66
HAWCH7-C4	1,538	0.0024	0.030	2.21	2.5 x 19.5	80
HAWCH7-C5	688	0.0070	0.030	3.10	2.5 x 19.5	80
HAWCH7-C6	1,284	0.0010	0.030	1.48	2.5 x 29	80
SHACH2-C1	1,649	0.0069	0.030	4.07	2.5 x 83.5	637
SHACH2-C2	1,310	0.0064	0.030	3.94	2.5 x 86.5	637
SHACH2-C3	965	0.0077	0.030	4.16	2.5 x 86.5	637
SHACH2-C4	2,264	0.0075	0.030	4.13	2.5 x 86.5	637
SHACH2-C5	2,783	0.0047	0.030	3.40	2.5 x 100	637
SHACH2-C6	500	0.0116	0.030	4.46	2.5 x 100	637
SHACH2-C7	714	0.0041	0.030	3.16	2.5 x 107.5	637
SHACH2-C8	1,252	0.0059	0.030	3.54	2.5 x 107.5	637
SHACH3-C1	863	0.0066	0.030	–	2.5 x 10	–
SHACH3-C2	830	0.0071	0.030	–	2.5 x 10	–
SHACH3-C3	546	0.0047	0.030	–	2.5 x 10	–
SHACH3-C4	2,030	0.0064	0.030	–	2.5 x 10	–
SHACH3-C5	1,667	0.0084	0.030	4.38	2.5 x 40	327
SHACH3-C6	2,104	0.0068	0.030	3.95	2.5 x 44.5	327
SHACH3-C7	454	0.0031	0.030	2.69	2.5 x 65	327
SHACH3-C8	500	0.0082	0.030	3.64	2.5 x 65	327
SHACH3-C9	1,306	0.0053	0.030	3.18	2.5 x 65	327
SHACH3-C10	503	0.0041	0.030	3.02	2.5 x 65	348
SHACH3-C11	484	0.0044	0.030	3.08	2.5 x 65	348
SHACH3-C12	891	0.0038	0.030	2.94	2.5 x 65	348

Label	Length (m)	Slope (m/m)	Manning's n	Velocity (m/s)	Size (m)	Flow (m^3/s)
SHACH3-C13	443	0.0022	0.030	2.32	2.5 x 97	421
SHACH3-C14	442	0.0042	0.030	2.81	2.5 x 97	421
SHACH3-C15	729	0.0031	0.030	2.75	2.5 x 124.5	641
SHACH3-C16	86	0.0010	0.030	1.59	2.5 x 267	795
SHACH4-C1	780	0.0040	0.030	3.13	2.5 x 94.5	554
SHACH4-C2	1,189	0.0053	0.030	3.40	2.5 x 94.5	554
SHACH4-C3	962	0.0026	0.030	2.55	2.5 x 116	554
SHACH4-C4	871	0.0076	0.030	3.51	2.5 x 116	554
SHACH4-C5	1,579	0.0037	0.030	2.83	2.5 x 116	560
SHACH4-C6	500	0.0066	0.030	3.38	2.5 x 116	560
SHACH4-C7	837	0.0036	0.030	2.81	2.5 x 116	560
SHACH4-C8	589	0.0054	0.030	3.19	2.5 x 116	562
SHACH4-C9	861	0.0026	0.030	2.54	2.5 x 118	562
SHACH4-C10	109	0.0010	0.030	1.59	2.5 x 253	753
SHACH5-C1	4,768	0.0055	0.030	3.35	2.5 x 20.5	127
SHACH5-C2	1,017	0.0081	0.030	3.77	2.5 x 20.5	127
SHACH5-C3	1,398	0.0034	0.030	2.78	2.5 x 46	239
SHACH5-C4	500	0.0066	0.030	3.42	2.5 x 46	239
SHACH5-C5	550	0.0040	0.030	2.93	2.5 x 46	239
SHACH5-C6	1,684	0.0031	0.030	2.66	2.5 x 50	249
SHACH6-C1	1,717	0.0040	0.030	2.76	2.5 x 14.5	75
SHACH6-C2	1,306	0.0014	0.030	1.78	2.5 x 46	154
SHACH6-C3	500	0.0080	0.030	3.06	2.5 x 46	154
SHACH6-C4	883	0.0030	0.030	2.26	2.5 x 46	154
SHACH6-C5	351	0.0060	0.030	2.81	2.5 x 46	154
SHACH6-C6	402	0.0082	0.030	3.09	2.5 x 46	154
SHACH7-C1	698	0.0070	0.030	3.61	2.5 x 15	99
SHACH7-C2	1,170	0.0042	0.030	3.03	2.5 x 29.5	167
SHACH7-C3	827	0.0075	0.030	3.62	2.5 x 29.5	167
SHACH7-C4	1,008	0.0031	0.030	2.64	2.5 x 43	211
SHACH7-C5	675	0.0036	0.030	2.77	2.5 x 43	211
SHACH7-C6	660	0.0017	0.030	2.01	2.5 x 56	211
SHACH7-C7	699	0.0039	0.030	2.86	2.5 x 56	273
SHACH7-C8	693	0.0028	0.030	2.54	2.5 x 57.5	273
SHACH7-C9	976	0.0027	0.030	2.52	2.5 x 58	273
SHACH8-C1	1,392	0.0058	0.030	3.22	2.5 x 13	76
SHACH8-C2	1,899	0.0030	0.030	2.43	2.5 x 17	76
SHACH8-C3	589	0.0062	0.030	3.06	2.5 x 17	76
SHACH8-C4	647	0.0021	0.030	2.06	2.5 x 20	76
SHACH8-C5	1,357	0.0039	0.030	2.52	2.5 x 20	76
SHACH9-C1	2,230	0.0055	0.030	2.98	2.5 x 10	53
SHACH9-C2	399	0.0052	0.030	3.29	2.5 x 22	136
SHACH9-C3	703	0.0010	0.030	1.52	2.5 x 47.5	136
SHACH9-C4	569	0.0071	0.030	2.78	2.5 x 47.5	136
SHACH9-C5	795	0.0048	0.030	2.46	2.5 x 47.5	136
SHACH9-C6	794	0.0010	0.030	1.53	2.5 x 52	149
SHACH10-C1	1,941	0.0061	0.030	3.60	2.5 x 25	168
SHACH10-C2	1,004	0.0016	0.030	1.95	2.5 x 57.5	210
SHACH10-C3	809	0.0084	0.030	3.23	2.5 x 57.5	210
SHACH10-C4	1,044	0.0033	0.030	2.45	2.5 x 57.5	213
SHACH10-C5	576	0.0026	0.030	2.48	2.5 x 72.5	336

Label	Length (m)	Slope (m/m)	Manning's n	Velocity (m/s)	Size (m)	Flow (m^3/s)
SHACH11-C1	1,352	0.0033	0.030	2.60	2.5 x 22	106
SHACH11-C2	1,075	0.0073	0.030	3.34	2.5 x 22	106
SHACH12-C1	649	0.0079	0.030	–	2.5 x 10	–
SHACH12-C2	800	0.0075	0.030	–	2.5 x 10	–
SHACH12-C3	1,076	0.0076	0.030	–	2.5 x 10	–
SHACH12-C4	501	0.0085	0.030	–	2.5 x 10	–
SHACH12-C5	748	0.0078	0.030	–	2.5 x 10	–
SHACH13-C1	1,642	0.0050	0.030	3.40	2.5 x 49.5	313
SHACH14-C1	1,856	0.0061	0.030	3.25	2.5 x 12	71
SHACH14-C2	653	0.0031	0.030	2.54	2.5 x 24	113
SHACH14-C3	500	0.0046	0.030	2.88	2.5 x 24	113
SHACH14-C4	935	0.0043	0.030	2.83	2.5 x 24	113
SHACH15-C1	821	0.0043	0.030	2.60	2.5 x 10	45
SHACH15-C2	659	0.0118	0.030	3.62	2.5 x 10	45
SHACH15-C3	843	0.0049	0.030	2.71	2.5 x 10	45
SHACH15-C4	559	0.0048	0.030	3.06	2.5 x 16.5	94
SHACH15-C5	500	0.0031	0.030	2.51	2.5 x 20	94
SHACH15-C6	1,205	0.0035	0.030	2.62	2.5 x 20	94
SHACH15-C7	809	0.0051	0.030	3.22	2.5 x 20	120
SHACH15-C8	1,412	0.0023	0.030	2.24	2.5 x 28.5	120
SHACH16-C1	1,505	0.0053	0.030	3.13	2.5 x 14	82
SHACH16-C2	500	0.0090	0.030	3.72	2.5 x 14	82
SHACH16-C3	667	0.0043	0.030	2.85	2.5 x 15.5	82
SHACH16-C4	1,050	0.0023	0.030	2.21	2.5 x 25.5	105
SHACH16-C5	499	0.0071	0.030	3.15	2.5 x 25.5	105
SHACH16-C6	1,887	0.0048	0.030	2.78	2.5 x 25.5	105
SHACH16-C7	317	0.0010	0.030	1.50	2.5 x 37.5	105
SHACH17-C1	1,762	0.0069	0.030	3.85	2.5 x 29	207
SHACH17-C2	970	0.0037	0.030	2.88	2.5 x 38.5	207
SHACH17-C3	500	0.0079	0.030	3.65	2.5 x 38.5	207
SHACH17-C4	1,172	0.0040	0.030	2.96	2.5 x 38.5	207
SHACH17-C5	704	0.0031	0.030	2.65	2.5 x 42	207
SHACH18-C1	733	0.0013	0.030	1.72	2.5 x 37	119
SHACH18-C2	500	0.0116	0.030	3.36	2.5 x 37	119
SHACH18-C3	537	0.0025	0.030	2.11	2.5 x 37	119
SHACH18-C4	1,250	0.0053	0.030	2.65	2.5 x 37	119
SHACH19-C1	1,812	0.0024	0.030	–	2.5 x 10	–
SHACH20-C1	858	0.0056	0.030	3.14	2.5 x 12.5	72
SHACH20-C2	1,243	0.0044	0.030	2.86	2.5 x 13.5	72
SHACH20-C3	1,040	0.0026	0.030	2.45	2.5 x 40.5	185
SHACH20-C4	614	0.0047	0.030	2.93	2.5 x 40.5	185
SHACH21-C1	1,089	0.0062	0.030	3.30	2.5 x 11.5	71
SARCH2-C1	681	0.0033	0.030	2.75	2.5 x 49	251
SARCH2-C2	1,037	0.0051	0.030	3.14	2.5 x 49	251
SARCH2-C3	517	0.0035	0.030	2.80	2.5 x 49	251
SARCH2-C4	799	0.0015	0.030	1.89	2.5 x 94	333
SARCH2-C5	497	0.0067	0.030	3.00	2.5 x 94	333
SARCH2-C6	2,611	0.0031	0.030	2.37	2.5 x 94	333
SARCH3-C1	3,014	0.0084	0.030	4.38	2.5 x 41	336
SARCH3-C2	760	0.0065	0.030	3.88	2.5 x 46.5	336
SARCH3-C3	2,179	0.0072	0.030	4.00	2.5 x 46.5	336

Label	Length (m)	Slope (m/m)	Manning's n	Velocity (m/s)	Size (m)	Flow (m^3/s)
SARCH3-C4	913	0.0058	0.030	3.66	2.5 x 49	336
SARCH4-C1	2,374	0.0030	0.030	2.36	2.5 x 15	65
SARCH4-C2	731	0.0087	0.030	3.33	2.5 x 15	65
SAKCH3-C1	1,858	0.0029	0.030	2.59	2.5 x 45	218
SAKCH3-C2	1,862	0.0023	0.030	2.31	2.5 x 60	259
SAKCH3-C3	702	0.0034	0.030	2.61	2.5 x 60	259
SAKCH3-C4	656	0.0026	0.030	2.41	2.5 x 60	259
SAKCH3-C5	979	0.0043	0.030	2.83	2.5 x 60	264
SAKCH3-C6	416	0.0010	0.030	1.56	2.5 x 90.5	264
SAKCH4-C1	1,348	0.0039	0.030	–	2.5 x 10	–
SAKCH4-C2	573	0.0015	0.030	–	2.5 x 10	–
SAKCH5-C1	1,677	0.0040	0.030	2.73	2.5 x 14	71
SAKCH5-C2	1,096	0.0010	0.030	1.53	2.5 x 52.5	149

APPENDIX B – PLAN AND PROFILE DRAWINGS

The plan and profile drawings of the 32 channels for the 200-year design are issued as a separate set; the sheets are listed by channel.

Channel	Sheets	First drawing	Last drawing
HAW-CH2-SS	9	2224-PD-HY-PP-B4-011	2224-PD-HY-PP-B4-019
HAW-CH3-SS	3	2224-PD-HY-PP-B4-020	2224-PD-HY-PP-B4-022
HAW-CH4-SS	3	2224-PD-HY-PP-B4-023	2224-PD-HY-PP-B4-025
HAW-CH5-SS	9	2224-PD-HY-PP-B4-026	2224-PD-HY-PP-B4-034
HAW-CH6-SS	3	2224-PD-HY-PP-B4-035	2224-PD-HY-PP-B4-037
HAW-CH7-SS	8	2224-PD-HY-PP-B4-038	2224-PD-HY-PP-B4-045
SHA-CH2-SM	13	2224-PD-HY-PP-B4-056	2224-PD-HY-PP-B4-068
SHA-CH3-SM	15	2224-PD-HY-PP-B4-069	2224-PD-HY-PP-B4-083
SHA-CH4-SM	10	2224-PD-HY-PP-B4-084	2224-PD-HY-PP-B4-093
SHA-CH5-SM	11	2224-PD-HY-PP-B4-094	2224-PD-HY-PP-B4-104
SHA-CH6-SM	7	2224-PD-HY-PP-B4-105	2224-PD-HY-PP-B4-111
SHA-CH7-SM	9	2224-PD-HY-PP-B4-112	2224-PD-HY-PP-B4-120
SHA-CH8-SS	7	2224-PD-HY-PP-B4-121	2224-PD-HY-PP-B4-127
SHA-CH9-SS	7	2224-PD-HY-PP-B4-128	2224-PD-HY-PP-B4-134
SHA-CH10-SS	7	2224-PD-HY-PP-B4-135	2224-PD-HY-PP-B4-141
SHA-CH11-SS	4	2224-PD-HY-PP-B4-142	2224-PD-HY-PP-B4-145
SHA-CH12-SS	5	2224-PD-HY-PP-B4-146	2224-PD-HY-PP-B4-150
SHA-CH13-SS	3	2224-PD-HY-PP-B4-151	2224-PD-HY-PP-B4-153
SHA-CH14-SS	5	2224-PD-HY-PP-B4-154	2224-PD-HY-PP-B4-158
SHA-CH15-SS	8	2224-PD-HY-PP-B4-159	2224-PD-HY-PP-B4-166
SHA-CH16-SS	8	2224-PD-HY-PP-B4-167	2224-PD-HY-PP-B4-174
SHA-CH17-SS	7	2224-PD-HY-PP-B4-175	2224-PD-HY-PP-B4-181
SHA-CH18-SS	5	2224-PD-HY-PP-B4-182	2224-PD-HY-PP-B4-186
SHA-CH19-SS	3	2224-PD-HY-PP-B4-187	2224-PD-HY-PP-B4-189
SHA-CH20-SS	5	2224-PD-HY-PP-B4-190	2224-PD-HY-PP-B4-194
SHA-CH21-SS	3	2224-PD-HY-PP-B4-195	2224-PD-HY-PP-B4-197
SAR-CH2-SM	8	2224-PD-HY-PP-B4-217	2224-PD-HY-PP-B4-224
SAR-CH3-SS	8	2224-PD-HY-PP-B4-225	2224-PD-HY-PP-B4-232
SAR-CH4-SS	5	2224-PD-HY-PP-B4-233	2224-PD-HY-PP-B4-237
SAK-CH3-SM	8	2224-PD-HY-PP-B4-263	2224-PD-HY-PP-B4-270
SAK-CH4-SS	3	2224-PD-HY-PP-B4-271	2224-PD-HY-PP-B4-273
SAK-CH5-SS	4	2224-PD-HY-PP-B4-274	2224-PD-HY-PP-B4-277

APPENDIX C – COSTS DASHBOARD

The estimate of 168.2 million OMR at a glance: the channel, dyke and culvert costs by catchment and by type, then the cost of every channel, dyke and culvert of the scheme.

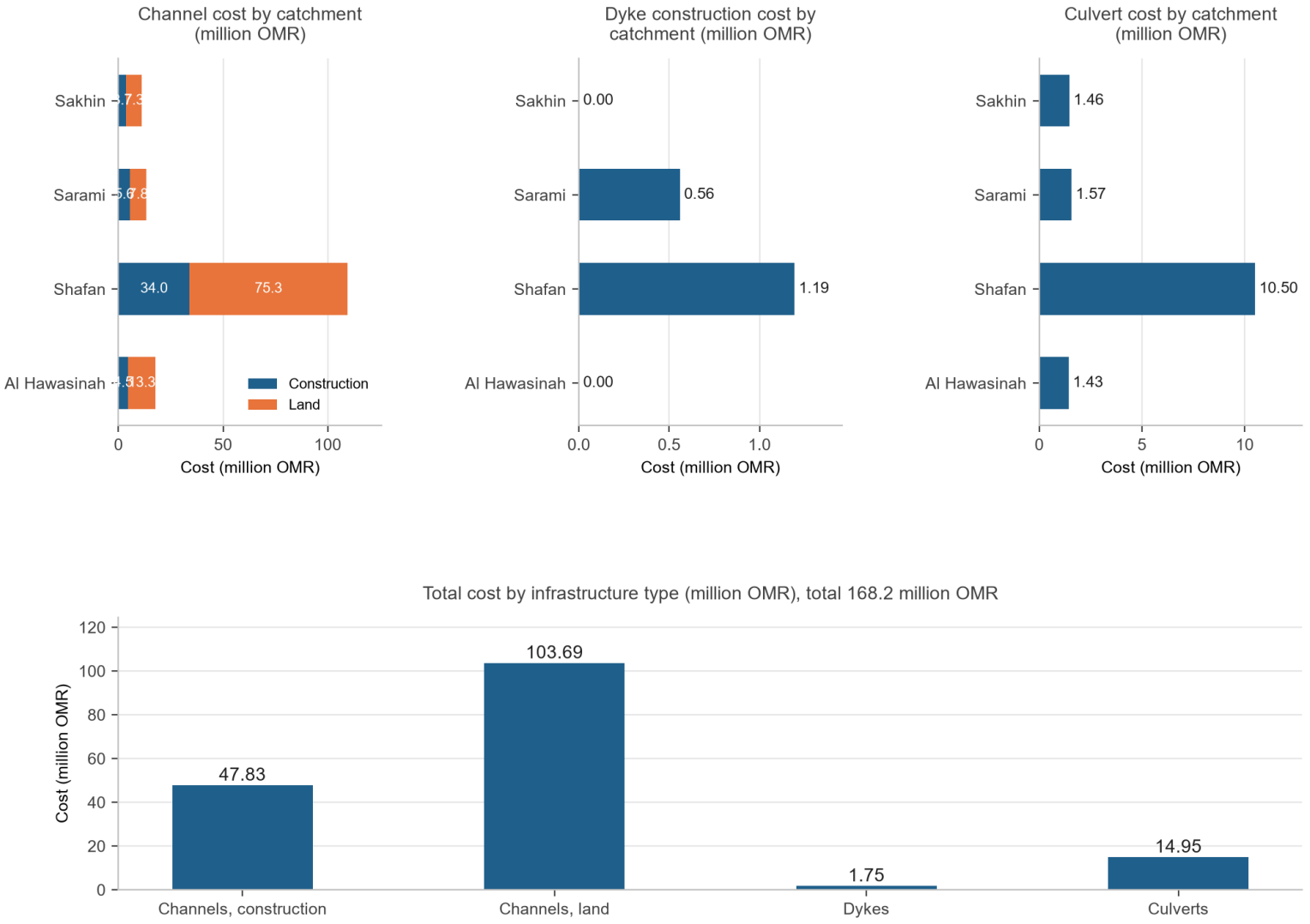


Figure 30 – Costs dashboard: channels, dykes and culverts by catchment, and the total by infrastructure type

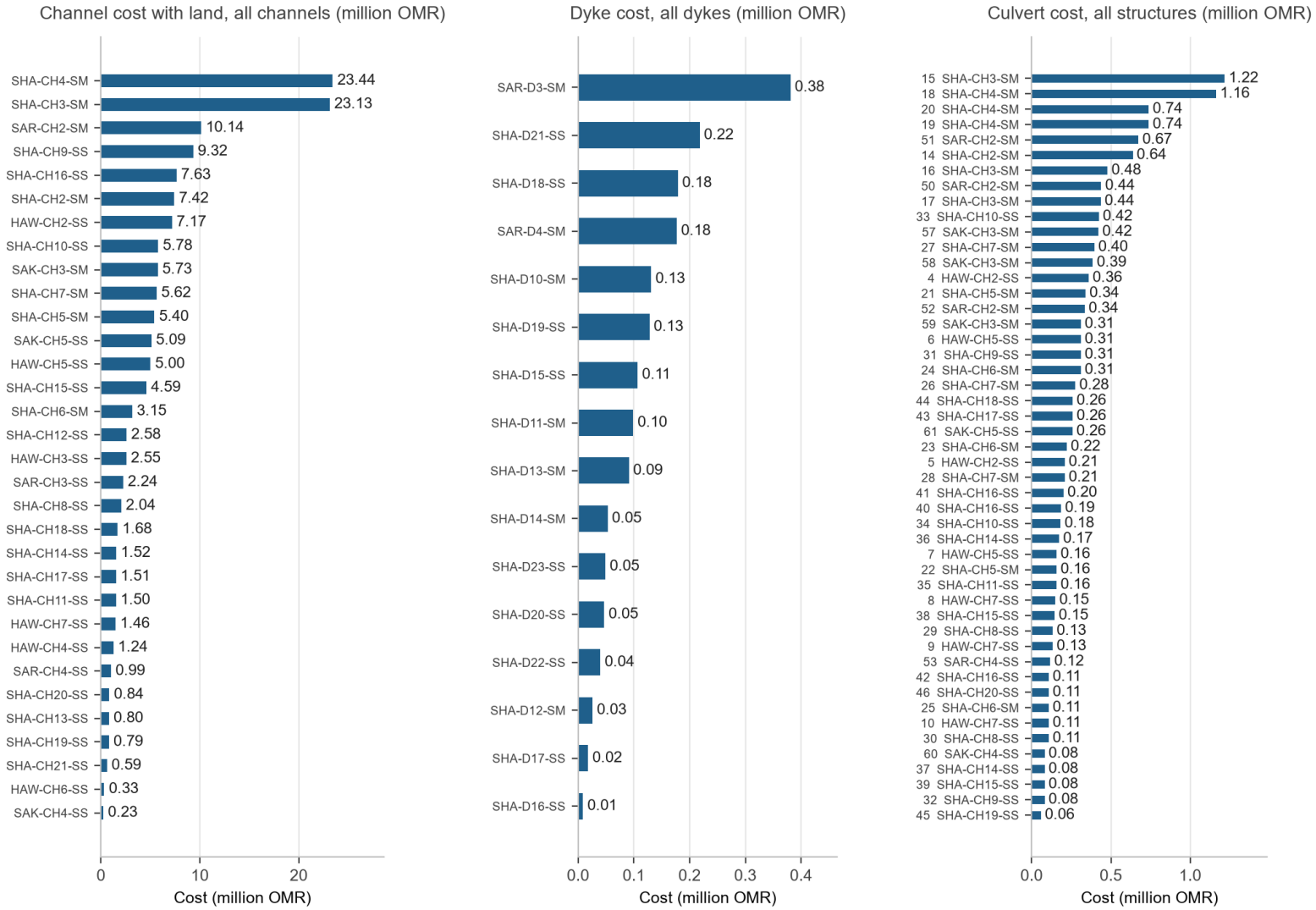


Figure 31 – Costs dashboard: every channel, dyke and culvert